



Advancing forest mapping: Pretraining strategies and deep-ensemble based uncertainty for predicting evergreen broad-leaved cover from Sentinel-2 time series

Benedikt Hiebl ^{a,*,} Nicola Alessi ^{b,} Giacomo Calvia ^{c,} Alessandro Bricca ^{c,} Gianmaria Bonari ^{d,e,} Giulio Zangari ^{c,} Wouter Dorigo ^{f,} Stefan Zerbe ^{c,g,} Martin Rutzing ^a

^a Department of Geography, University of Innsbruck, Innsbruck, Austria

^b Italian Institute for Environmental Protection and Research, Rome, Italy

^c Faculty of Agricultural, Environmental and Food Sciences, Free University of Bozen-Bolzano, Bozen, Italy

^d Department of Life Sciences, University of Siena, Siena, Italy

^e National Biodiversity Future Center NBFC, Palermo, Italy

^f Department of Geodesy and Geoinformation, Technical University of Vienna, Vienna, Austria

^g Institute of Geography, University of Hildesheim, Hildesheim, Germany

ARTICLE INFO

Dataset link: <https://benedikthiebl.users.earthengine.app/view/test>, https://git.uibk.ac.at/c7161037/traceve_pretraining, <https://doi.org/10.5281/zenodo.15000771>

Keywords:

Forest ecosystems
Satellite remote sensing time series
Sentinel-2
Transfer learning
Uncertainty

ABSTRACT

The distribution changes of evergreen broad-leaved tree and shrub species (EVE) at the border of Mediterranean and temperate forests due to climate change and land-use changes necessitates accurate mapping techniques to support biodiversity monitoring and climate adaptation strategies. Remote sensing time series provide valuable data for vegetation analysis, yet functional-level mapping in mixed forests remains challenging due to limited observation data. To tackle this limitation, the present study investigates pretraining strategies for mapping EVE cover in selected Italian forest areas using Sentinel-2 time series and a probabilistic Convolutional Neural Network. Additionally deep ensemble-based uncertainty estimation is used to further enhance the interpretability of the output. We compare three model training strategies: (i) direct training on up-to-date but limited field data, (ii) supervised pretraining on a larger, diverse forest vegetation database before fine-tuning, and (iii) self-supervised pretraining on large-scale unlabeled time series before fine-tuning. Our results demonstrate that pretraining on contextually similar datasets in combination with a spatial split of training and validation data significantly enhances predictive performance and generalization to unseen regions in a cross-validation experiment. Additionally, we assess epistemic and aleatoric uncertainty to improve interpretability and identification of out-of-distribution predictions. This study highlights the benefits of pretraining and uncertainty quantification for large-scale remote sensing applications with limited availability of labeled data.

1. Introduction

The redistribution of climatic drivers throughout the twentieth century has led to significant shifts in tree species ranges and growth limitations globally (Babst et al., 2019). In parts of Europe, particularly in southern regions, these changes have driven a transition from cold-adapted deciduous forests to Mediterranean ecosystems. In these areas evergreen broad-leaved species (EVEs) are expanding due to milder winter temperatures, forest regrowth following land abandonment, and rising atmospheric CO₂ levels (Peñuelas and Boada, 2003; Conedera et al., 2017, 2018). Understanding and monitoring these shifts is essential for biodiversity conservation and climate adaptation strategies,

yet supra-regional-scale assessments of EVEs cover and diversity remain scarce.

Remote sensing (RS) offers a powerful tool for tracking biodiversity trends, providing spatially and temporally continuous data to analyze ecosystem changes (Pettorelli et al., 2018). Freely available multispectral time series from Sentinel-2/3 and Landsat satellites have been widely used for vegetation monitoring, forest cover mapping, and tree species classification (Bae et al., 2019; Bolyn et al., 2022). However, distinguishing functional-level patterns in mixed forest stands remains a key challenge, given the complex spectral signatures and seasonal dynamics of EVEs, deciduous broad-leaved species and evergreen

* Corresponding author.

E-mail address: Benedikt.Hiebl@uibk.ac.at (B. Hiebl).

<https://doi.org/10.1016/j.jag.2025.104734>

Received 31 March 2025; Received in revised form 9 July 2025; Accepted 11 July 2025

Available online 24 July 2025

1569-8432/© 2025 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

needle-leaved tree and shrub species. Functionally distinct species often exhibit similar spectral reflectance during overlapping phenological phases, hindering separability in single-date imagery. Dense Sentinel-2 time series can capture intra-annual phenological dynamics, which improves tree species-level classification performance (Chabalala et al., 2023; Grabska et al., 2019). However, species-level variability in green-up and senescence timing can introduce uncertainty. Higher spatial and spectral resolution has shown to enhance discrimination, particularly for non-dominant or underrepresented species in structurally complex canopies (Immitzer et al., 2012; Schiefer et al., 2020; Sylvain et al., 2024).

Recent advancements in machine learning (ML) and deep learning (DL) have shown promise in overcoming these limitations, delivering high classification accuracy even with imbalanced and heterogeneous training datasets (Muro et al., 2022; Praticò et al., 2021). Applying DL to RS vegetation analysis often faces a critical bottleneck: the scarcity of high-quality labeled data for training. Addressing this small data problem requires leveraging novel pretraining strategies, transfer learning, or self-supervised learning techniques to improve model generalization and robustness in species-level mapping (Safonova et al., 2023).

Critical aspects in DL for forestry applications include (i) the amount of required training data (often from national inventories or RS platforms), (ii) training data diversity, and (iii) the model's ability to capture temporal dynamics (Hamedianfar et al., 2022). Many studies focus on target variables easily observed by RS platforms, such as above-ground biomass measured by GEDI (Lang et al., 2023; Dubayah et al., 2020) or forest cover type using optical imagery (Li et al., 2023). Other studies focus on areas with national forest inventories or consider only pure stands (Bolyn et al., 2022). Only 22% of studies combining RS, forestry, and DL used in-situ observations for training and validation (Kattenborn et al., 2021), mostly due to its discrete spatial nature and sparse data availability. They further state that decreasing the training data size shows significant performance decrease across several studies and across different model architectures. Plot-wise observations also pose problems due to the 1D nature of the spectral signals at resolutions similar to plot size, making them incompatible with image-based CNNs. Therefore less established model architectures specifically for time series extrinsic regression (TSER) or classification (TSC) are required (Tan et al., 2021). Nevertheless, high-quality, time-sensitive observations for biodiversity mapping or specific mapping tasks can only be recorded in small plots. In this context, recent work showed that pretrained Vision-transformers improve accuracy on plot- and pixel-level mapping (Ge et al., 2023), and under biome shifts (Fuller et al., 2022). Other research areas show that transfer learning (TL) enhances prediction robustness in out-of-distribution (OOD) scenarios when used in similar contexts with scarce training data availability (Liu et al., 2023). However, studies investigating the specific challenges arising from combining satellite remote sensing time series (STS), scarce plot observations and DL remain rare (Safonova et al., 2023).

Studies show that TL improves performance in vegetation monitoring only for applications with scarce training data, although most focus solely on prediction accuracy metrics and do not consider generalizability on out-of-distribution or corrupted data input (Kattenborn et al., 2021). In contrast, Hendrycks et al. (2019) demonstrated that pre-training benefits both robustness and uncertainty, especially interesting in large-scale mapping from limited training data. Many TL studies focus on satellite imagery or satellite image time series, suited for 2D- or 3D-CNNs, respectively. For detailed vegetation plot observations smaller or equal to the sensor's ground-sample distance, a 1D spectral-temporal approach on pixel-level time series is necessary, which is underinvestigated in RS with respect to TL. To the best of our knowledge, this is the first study combining TL, in-situ plot observations, and 1D TSER for estimating EVE cover in ecosystems where EVEs are often non-dominant species. Our approach addresses the specific challenges of functional-level mapping in mixed forests under limited plot observations. We employ context similarity pretraining, where the

model is pretrained on RS data with spectral-temporal characteristics closely matching our target regions.

Measures of uncertainty in DL model predictions are widely used in decision-sensitive applications such as medical diagnosis (Pocevičiūtė et al., 2022). In RS applications only few studies account for prediction uncertainties (Lang et al., 2023; Sylvain et al., 2024). Models facing sparse training data, OOD data or a domain shift (Ma et al., 2024) - common in large-scale mapping from in-situ observations - uncertainty can help describing model generalizability and prediction quality (Tom et al., 2023). Validating against reference datasets typically serves the purpose of testing prediction accuracy, rather than uncertainty analysis of the retrieval model (Tran et al., 2023). In this context, uncertainty quantifies the models performance against an unknown true value. In areas lacking reference data, epistemic uncertainty (EU) serves as a measure of model prediction confidence and enhances product interpretability.

This study investigates STS-based DL strategies for large scale mapping of forest EVE fractional cover in Temperate and Mediterranean forests under limited training data availability. Therefore the objectives of this study are:

- evaluating three different strategies for model training regarding generalization capability and prediction uncertainty
- assessing the value of prediction uncertainty measures for accuracy and interpretability

In this specific study setting, generalization refers to (i) the prediction performance in spatially different regions with unknown species composition, and (ii) the prediction performance on OOD samples, particularly for areas with needle-leaved evergreen tree cover. Both are highlighting challenges tied to spatial heterogeneity and distributional shifts in RS data. This study builds on established ML-based vegetation mapping techniques across diverse sensors, scales, and algorithms (Adagbasa et al., 2022; Guo et al., 2020; Ho Tong Minh et al., 2018; Kacic et al., 2021; Kollert et al., 2021). Sentinel-2 L2A time series and derived vegetation indices, capturing the phenological cycle of different plant species and widely used in vegetation RS applications due to its high spectral, spatial, and temporal resolution (Dobrinčić et al., 2022; Hemmerling et al., 2021; Grabska et al., 2019), serve as predictors in Convolutional Neural Networks (CNNs) mapping EVE cover across distinct forest areas in Italy.

2. Materials and methods

2.1. Study area

The study uses data from five forested areas situated in Italy, exhibiting distinct forest composition (Fig. 1). Sibillini National Park (SIB) and Gran Sasso National Park (GRA), located within the Apennines in Central Italy (350–2912 m a.s.l.), are dominated by mountainous temperate forest. The prevalent deciduous tree species is common beech (*Fagus sylvatica* L.). In contrast, the Mediterranean forest in Gennargentu Park (GEN), located on the eastern coast of Sardinia (0–1834 m a.s.l.), is characterized by the dominance of holm oak (*Quercus ilex*). Cilento National Park (CIL) is the most diverse with Mediterranean vegetation in the coastal regions and sub-Mediterranean beech forest inland. Nebrodi Regional Park (NEB) on Sicily features forests with evergreen and deciduous oaks or beech in higher elevations. Despite the occurrence of mixed deciduous and EVE forest stands in all areas, their distribution varies with elevation, latitude, and climatic conditions. These factors result in a high presence of EVE species in GEN, CIL, and NEB, and a predominance of deciduous species in SIB and GRA. The five areas unique forest vegetation characteristics enable testing model generalization across different forest types and distant regions. As this study focuses on forested areas only, non-forest areas were masked using the Copernicus Forest Type Classification (FTY) (Langanke et al., 2018).

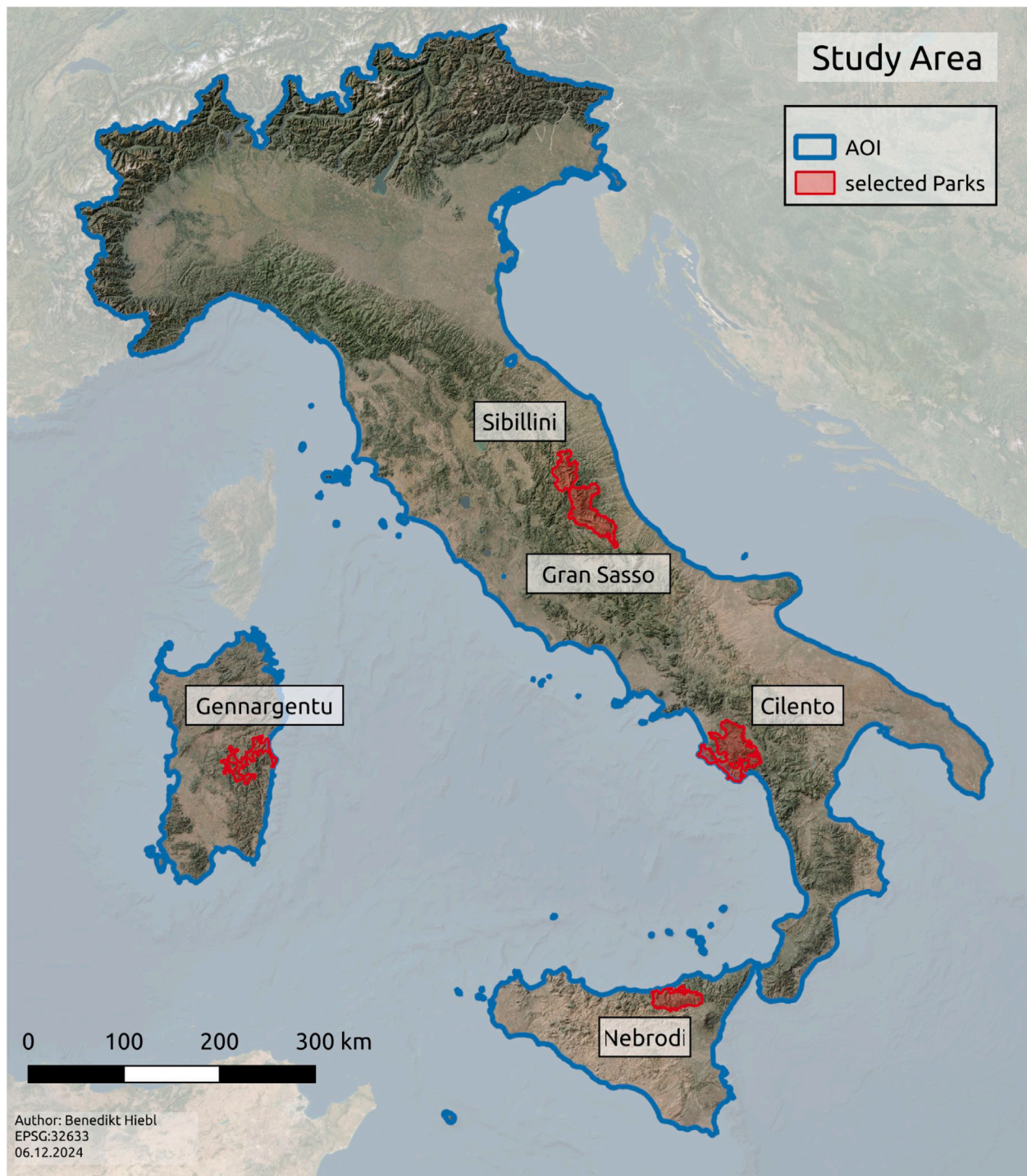


Fig. 1. Location of the five training and test areas in the cross-validation scheme in Italy.

2.2. Methodological overview

The present study explores three training strategies for a DL model to address the limited amount of high-quality training data. After collecting and preparing field survey and database data, the workflow further includes STS preprocessing, feature selection, (pre-)training modified InceptionTime DL models, mapping EVE cover from Sentinel-2 L2A time series, and finally estimating model robustness and generalizability using validation data and uncertainty estimates (Fig. 2). For comparison, a Random Forest model was trained on the same set of features.

2.3. Datasets

2.3.1. Forest plot observations

The dataset (VPO2024) comprises 360 site-level woody vegetation plot observations sampled and balanced across 5 forested areas (Fig. 3). To consider all the forest vegetation belts of the areas, the sampling scheme was based on elevation (10 m DEM, Tarquini et al., 2012) and forest cover (at least 80% of broad-leaved forest), subdivided into 4 elevational bands delimited by quartile values of mean cell elevation. 6 cells were then randomly selected within each elevational band. Each of the sampled cells was then divided into two elevational bands delimited

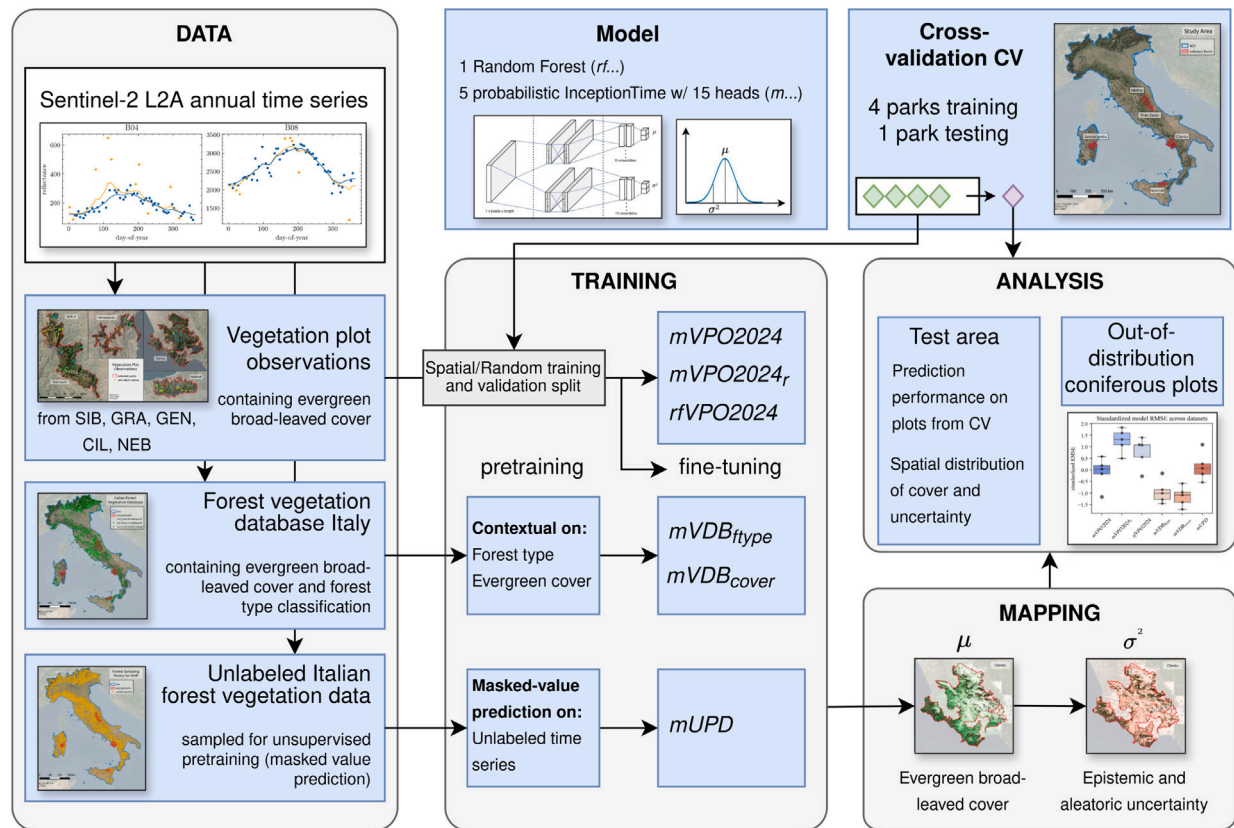


Fig. 2. Workflow of the study. First involving the combination of RS time series and different forest vegetation plot datasets. Secondly training probabilistic models in a cross-validation scheme using different training strategies and datasets. The concluding analysis involves mapping of the test areas and testing the models performances.

at the median elevation values of the cell and a site of 900 m² was randomly selected for each cell elevational band. A further preferential site targeted to EVEs-rich forest stands was selected to account for EVEs limited distribution with respect to deciduous species (Alessi et al., 2023). Each site is composed of 4 systematic sampling plots of 100 m² for a total of 1440 sampling plots, where species-level ground cover estimation was recorded across three vertical layers: 0 m to 1 m; 1 m to 10 m; 10 m to maximum height. The EVE cover target was obtained by aggregating cover from the upper two layers at site level. Spatial autocorrelation between the individual sampling plots was countered by clustering the sites into 20 spatially distinct regions (4 for each area) with equal sample size (between 71 and 73 samples) using a constrained kmeans-clustering based on latitude, longitude, and park (Fig. 4) (Bradley et al., 2000). The training and validation split of the dataset was based on the clusters and not from a random sampling, while the test split was based on the area. Regarding the test sites, SIB and GRA are adjacent, with a minimum distance of 2.1 km between the closest test and training plot observation. For all other parks, the closest distance is 239 km (between GRA and CIL). The clustering-based train-validation split did not enforce a minimum distance constraint. Inter-cluster distance averaged 8.7 km (min 307 m), with 88% of plots (1,276 observations) having a nearest neighbor in a different cluster farther than 2 km.

2.3.2. Forest vegetation database

The dataset for pretraining in the TL approach is a collection of plot observations from different studies across Italy (Forest vegetation Database Italy, VDB) (Alessi et al., 2019). The dataset covers studies in all regions of Italy with different objectives, sampling strategies, and from 1972 until 2017 (Fig. 5). Although the quality of each plot observation cannot be estimated, the dataset represents a more diverse distribution across Italian forests as the reference dataset conducted

within this study. Therefore it can be assumed that pretraining the model on this dataset increases the robustness and generalizability due to the larger variability in spectral signatures within the dataset. The whole dataset provides 16,908 plot observations. Plots conducted earlier than 2000 or with no valid EUNIS forest habitat classification (Chytrý et al., 2020) were excluded. Although Sentinel-2 data is only available from 2016 onwards, the assumption is, that species-level changes of cover or abundance might occur within the plots, but do not change the forest type, assuming that the models can learn useful overall patterns in the data during pretraining. The two final datasets for supervised pretraining consist of 4807 plots (i) classified into forest types (*VDB_{ftype}*), and (ii) evergreen broad-leaved, deciduous broad-leaved and evergreen needle-leaved cover (*VDB_{cover}*).

2.3.3. Sampling for self-supervised learning

For self-supervised learning, 100,000 points (UPD) were sampled from all forested areas in Italy (masked based on FTY) (Fig. 5). The points were stratified using k-means clustering on 2023 30m resolution Sentinel-2 L2A summer/winter means and latitude, yielding 30 equally sized clusters. A balanced random sampling was then used for each cluster to retrieve an equal amount of final points.

2.3.4. Sentinel-2 L2A time series

For each plot location in the VPOs, VDB, and UPD the full Sentinel-2 L2A time series for the corresponding pixel starting with 01.01.2017 and ending with 31.12.2023 was retrieved. Sen2Cor Scene Classification (SCL) masks were applied to retain only data that is labeled “not-vegetated” or “Vegetation”. Finally vegetation indices (see A) were calculated for each time series and added to the feature set. Each of the three datasets finally consists of a point Plot ID with distinct location, a preprocessed Sentinel-2 L2A time series and target variables in case of VPOs and VDB. These operations were performed

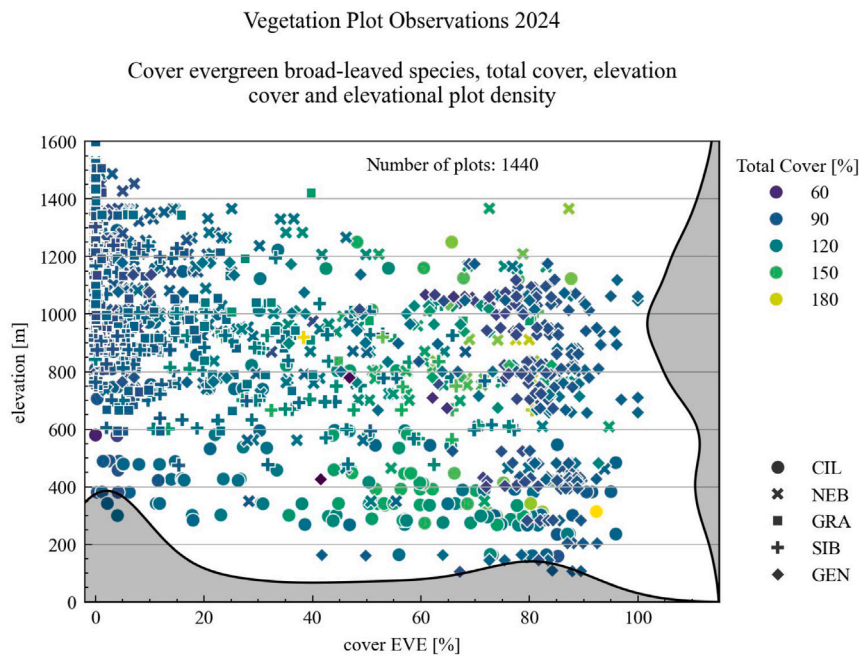


Fig. 3. Distribution of VPO2024 across EVE cover values (in %), elevation and parks. There is a gap in medium cover values between 20%_{cover} and 70%_{cover} with less observations. Total cover (in %) refers to the accumulated evergreen broad-leaved and deciduous cover in all vertical layers of the plot, which can exceed 100%_{cover}.

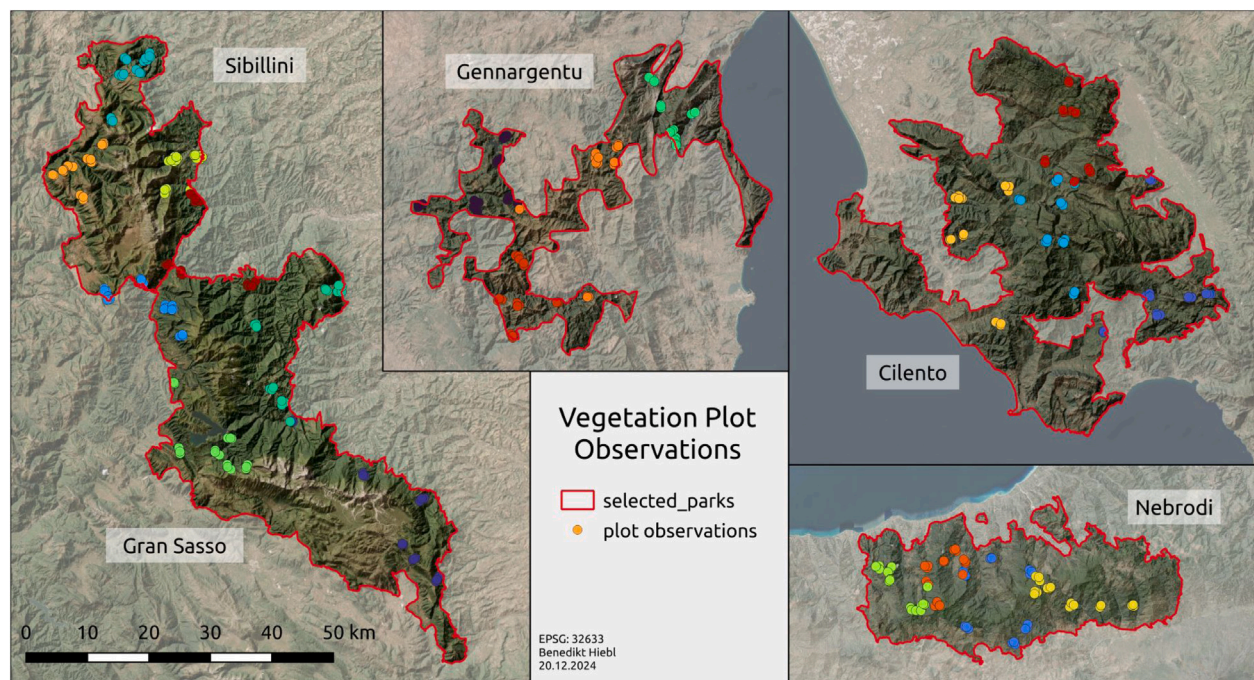


Fig. 4. Detailed view of the five investigated National Parks in Italy (cp. Fig. 1) containing spatially clustered Vegetation Plot Observations.

on Microsofts Planetary Computer Hub, which allowed fast data access and computation close to the data storage.

To reduce potential overfitting on noise in the data, especially in small datasets (Theunissen et al., 2019; Zhang et al., 2021), a standard preprocessing workflow for STS was applied. A three year time series (2021 - 2023) in the VPO2024 dataset was aggregated to a single synthetic year and resampled to a three day interval to reduce gaps in the observed annual phenological cycle. Denoising the STS consisted of: (i) outlier removal using Inter-quartile Range (IQR), to eliminate residual

outliers still abundant after SCL masking, and (ii) temporal smoothing using Whittaker Smoothing Algorithm, to interpolate missing values and smooth the noisy STS for phenology reconstruction (Atkinson et al., 2012; Khanal et al., 2020; Shao et al., 2016) (Fig. 6). Vegetation indices were computed after preprocessing to prevent noise and outlier propagation to the index time series. As last step before training the CNNs the data was normalized per band using the 2nd and 95th percentile respectively (Bolyen et al., 2022). For pretraining on the VDB datasets the full S2 time series (2017 to 2023) was split into annual

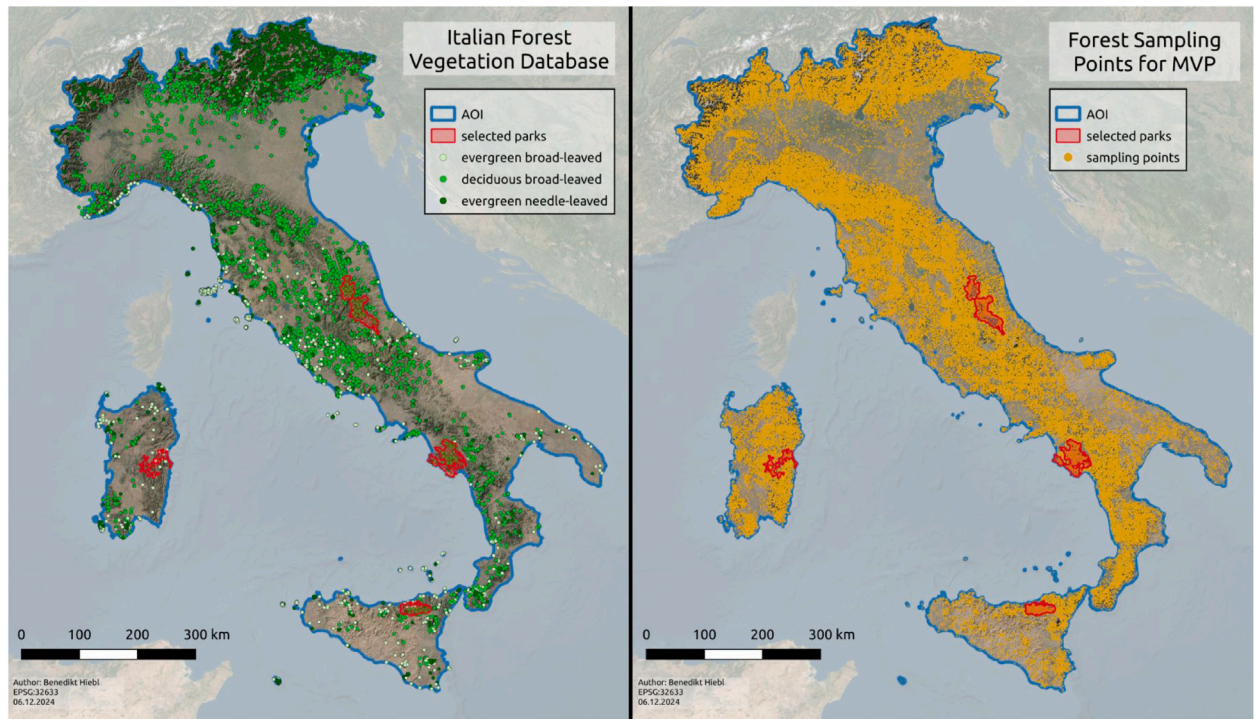


Fig. 5. (Left) Italian Forest Vegetation Database plot locations classified into leaf phenology and morphology types. (Right) Randomized distribution of plot locations (UPD) for Sentinel-2 time series retrieval for unsupervised masked value prediction (MVP).

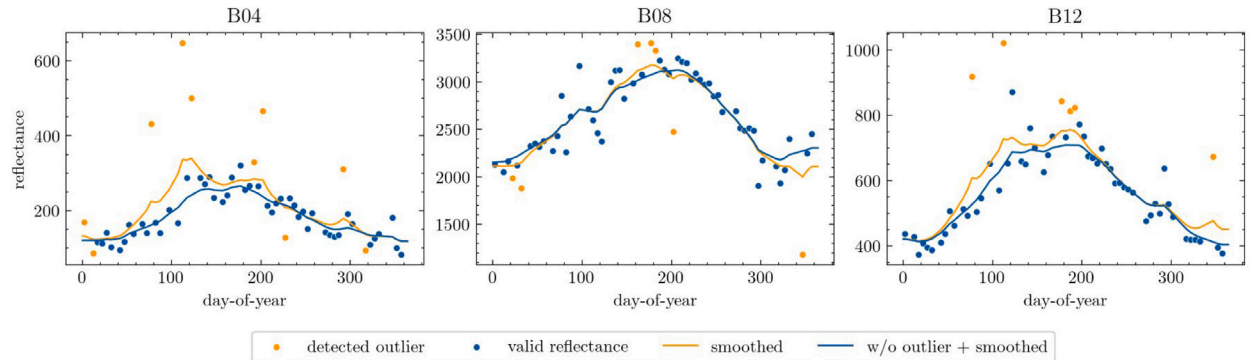


Fig. 6. Aggregated time series (2022 and 2023) of Plot "15_gen" in GEN with 90%_{cover} EVE at different S2 bands. Orange dots indicate outliers that are detected by IQR, while blue dots indicate valid observations. The orange and blue line shows the discrepancy between smoothing the annual time series based on the raw data versus based on the outlier-reduced data, affecting the spectral-phenological response of the signal.

cycles, as the models are trained on annual time series. This step serves as a kind of non-random data augmentation as it extends the training data size and introduces inter-annual variability of the phenological cycle despite the same class labels or cover values, respectively. The VDB datasets therefore contain 33649 data points for pretraining.

2.4. Deep learning model

2.4.1. Model architecture

We modified a CNN based on the InceptionTime architecture (Ismael Fawaz et al., 2020) to predict a continuous target variable (EVE cover). Sentinel-2 L2A spectral bands and their derivatives were treated as multivariate time series ($n_{samples}, n_{bands}, n_{timesteps}$). Each convolutional inception block (CIB) applies temporal filters independently across spectral bands, enabling the model to learn band-specific and joint

phenological patterns. We opted for a CNN-based model architecture, although state-of-the-art transformer models often outperform CNNs in many applications, but their high data demands and risk of overfitting on small datasets or noise make a CNN more suitable for our case (Kiefer et al., 2024; Rather et al., 2024). Additionally the parallel spectral-temporal convolutions and different kernel sizes, which reflect the receptive field of the InceptionTime model, can be easily adapted to sequences of any length, spectral bands, and specific needs, making adaptation of the model to other tasks easy. To estimate predictive uncertainty, we implemented, (i) a learned variance to model aleatoric (model-wise) uncertainty (AU) (Lakshminarayanan et al., 2017), and (ii) as measure of epistemic (data-wise) uncertainty (EU) the variance of ensemble predictions. Therefore we exchanged the model head after the CIB with an ensemble of 15 two-layer fully-connected networks, computing variance across cover predictions as measure of EU. To generate a probabilistic behavior of the network two models of the

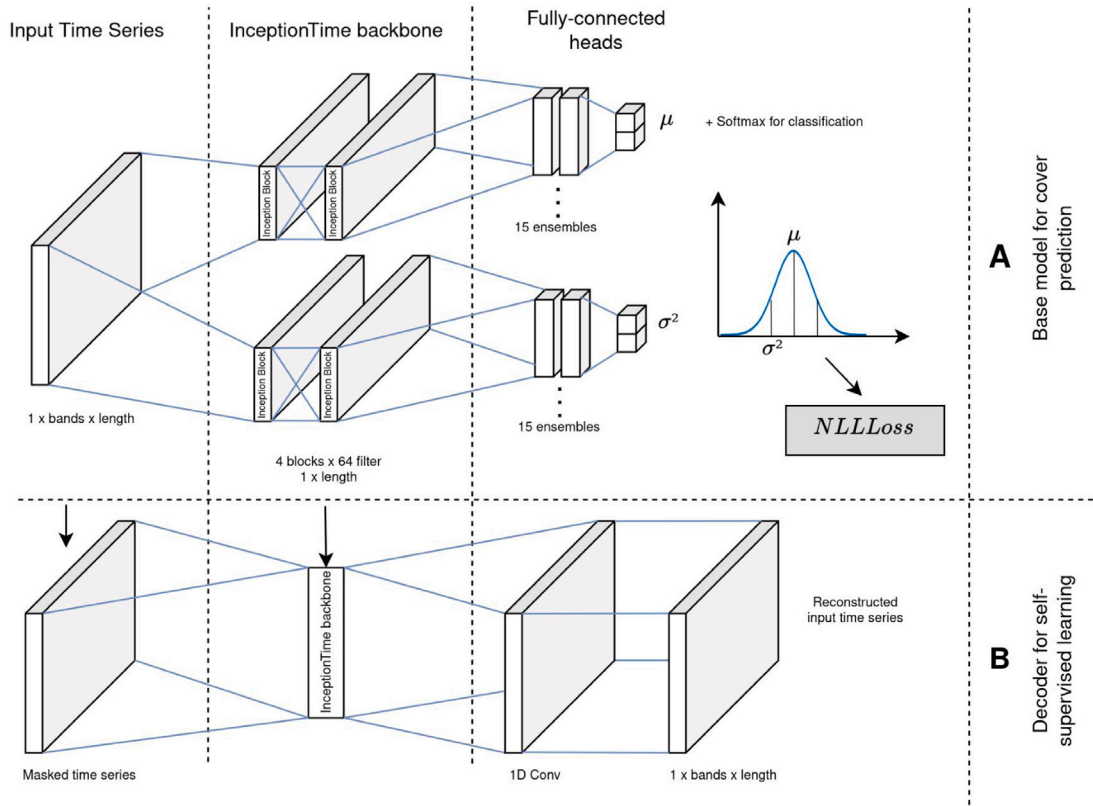


Fig. 7. Model architectures for $mVPO2024$, $mVDB_{ftype}$, $mVDB_{cover}$ (A) and $mUPD$ (B).

same architecture were used, one learning the predicted mean $\mu(x)$ and the second a variance $\sigma^2(x)$ as estimation for a gaussian distribution of model predictions and measure of AU. Separating prediction of $\mu(x)$ and $\sigma^2(x)$ in two unrelated models proved to have a more stable learning process than using one model for both, at the disadvantage of higher computational cost (Fig. 7). A Gaussian Negative-Log Likelihood Loss ($\mathcal{L}_{NLL}$), that treats the observed values as a sample from a gaussian distribution (Lakshminarayanan et al., 2017; Nix and Weigend, 1994), was used in an Adam optimizer to update model weights (Eq. (1)).

$$\mathcal{L}_{NLL}(y_n|x_n) = \frac{1}{2} \log(\sigma^2(x)) + \frac{(y - \mu(x))^2}{2\sigma^2(x)} \quad (1)$$

where y is the target variable. This architecture is used for fine-tuning the model on the VPOs. The model was implemented in PyTorch following the unofficial implementation by Oguiza (2023). The CIB serves as feature extractor for the supervised TSER task and as encoder in the MVP pretraining. The MVP decoder consists of a simple Multilayer Perceptron that outputs the reconstructed time series.

2.4.2. Training procedure

Hyperparameter tuning, training and inference were performed on a NVIDIA A30 GPU at the LEO5 High-Performance Computer at the University of Innsbruck. The training process was executed in four steps: (1) optimizing the method for preprocessing the time series data and reducing the amount of features, (2) optimizing and training the model on the $VPO2024$ (model name: $mVPO2024$), (3) training three distinct models with the same architecture on VDB_{ftype} ($m_{base}VDB_{ftype}$), VDB_{cover} ($m_{base}VDB_{cover}$) and UPD ($m_{base}UPD$) using a TL approach, (4) finetuning the pretrained models on the $VPO2024$ ($mVDB_{ftype}$, $mVDB_{cover}$ and $mUPD$). Additionally to $mVPO2024$, which was trained on spatially split $VPO2024$ dataset, $mVPO2024_r$ has been trained on a random split of the data.

In the first case, supervised pretraining involved a cross-task setup, where the pretraining task differs from the finetuning objective. The model $m_{base}VDB_{ftype}$ classifies the input STS features into distinct forest classes, while the finetuning involves a regression problem on EVE cover. In the second case, $m_{base}VDB_{cover}$ serves as a regression model, that predicts an evergreen broad-leaved cover value, which is close to the finetuning target (EVE cover). We assume that higher task similarity between pretraining and finetuning enhances the transfer effectiveness (Li et al., 2022b; Bernico et al., 2019) and ideally generalizability. In our self-supervised training approach the main goal is to enable the model to learn meaningful representations of the input time series. As a major problem within the STS are invalid observations (clouds, snow) during winter, the model is supposed to be able to reconstruct the phenology (or phenological features) from the available observations, despite the total absence of data during these periods. Therefore a data corruption procedure is introduced into the training process, that corrupts the training data during certain periods of the STS by setting the values to Nodata. An unsupervised learning technique called Masked Value Prediction (MVP) is used to enhance the models capability to learn meaningful features from the STS (Zerveas et al., 2020). We opted for MVP rather than other self-supervised algorithms, such as contrastive learning (Li et al., 2022a), because MVP leverages temporal continuity and structural phenological patterns commonly found in vegetation STS. The objective is to reconstruct temporally coherent reflectance curves. RS data exhibit structured missingness due to clouds or snow, which is not fully captured by random masking. To address this, our feature synchronous masking strategy involved a mean mask window length (16% of the year) and a mean total amount of masked data (25%) to make sure the reconstruction task matches the problem of typical missing data in STS, although we did not consider seasonality of masking. During training, the masked STS runs through the CIB backbone into a single layer 1D convolutional model head, that

reconstructs the original STS. Backpropagation is performed using a Mean Squared Error loss, computed between the reconstructed parts of the predicted spectral-temporal signature and the corresponding original STS. For fine-tuning a pretrained model several approaches are possible (Abdalla et al., 2019; Ma et al., 2024; Yosinski et al., 2014). In this study the CIB encoder weights were frozen during the finetuning process, to preserve the learned features in the downstream task. Each of the three pretrained models is then finetuned on the VPO dataset for EVE cover prediction using the same architecture and $\mathcal{L}_{\text{NLL}}$ loss. In comparison, non-pretrained models were directly trained on VPO with randomly initialized model weights.

To disentangle epistemic and aleatoric uncertainty, we used a deep ensemble architecture with a shared CIB followed by 15 independently initialized prediction heads. EU was estimated as the variance across these EVE cover ensemble predictions, capturing model uncertainty due to limited data availability or knowledge gaps. In contrast, AU was estimated using a separate CIB and heads trained to predict per-pixel variance that is combined with EVE cover in $\mathcal{L}_{\text{NLL}}$, reflecting observation noise and inherent data ambiguity. To ensure sufficient diversification for ensemble-based EU estimation, a separate dataloader is constructed for each model head using the full dataset. Each dataloader batch is first processed by the CIB feature extractor before being forwarded to its corresponding head (Algorithm 1). This approach in combination with individual weight initialization and dropout rate ensures that each of the ensemble members achieves the necessary diversity to effectively capture EU (Lakshminarayanan et al., 2017; Lang et al., 2023; Yang and Yee, 2024). In this context EU specifically arises from prediction variance, while AU arises from observational and labeling noise.

As a baseline and additional comparison, a Random Forest model was trained on the $VPO2024$ ($rVPO2024$). As this type of model cannot directly work with the two dimensions of spectral-temporal features the dataset was flattened to a 1D tensor and therefor lost its temporal dependencies. Also an uncertainty estimate for the prediction cannot be provided.

Algorithm 1 Estimate ensemble mean (EM), Epistemic (EU), and Aleatoric Uncertainty (AU)

Require: Dataset $D = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^N$, ensemble size M

- 1: Initialize shared backbones CIB_{cover} , CIB_{ale}
- 2: **for** $m = 1$ to M **do**
- 3: Initialize heads $H_{\text{cover},m}$, $H_{\text{ale},m}$ with different random seed
- 4: Create dataloader $\mathcal{D}_m$ with different shuffle
- 5: **for all** $(\mathbf{x}_i, \mathbf{y}_i) \in \mathcal{D}_m$ **do**
- 6: $\hat{\mu}_i^{(m)} \leftarrow H_{\text{cover},m}(CIB_{\text{cover}}(\mathbf{x}_i))$
- 7: $\hat{\sigma}_i^{(m)} \leftarrow H_{\text{ale},m}(CIB_{\text{ale}}(\mathbf{x}_i))$
- 8: $\mathcal{L}_{\text{NLL}} \leftarrow \frac{1}{2} \log \left(\hat{\sigma}_i^{(m)} \right) + \frac{(\mathbf{y}_i - \hat{\mu}_i^{(m)})^2}{2\hat{\sigma}_i^{(m)2}}$ ▷ Loss
- 9: Update using backpropagation
- 10: **end for**
- 11: **end for**
- 12: $\bar{\mu}_i \leftarrow \frac{1}{M} \sum_{m=1}^M \hat{\mu}_i^{(m)}$ ▷ EM
- 13: $AU_i \leftarrow \frac{1}{M} \sum_{m=1}^M \hat{\sigma}_i^{(m)2}$ ▷ AU
- 14: $EU_i \leftarrow \frac{1}{M} \sum_{m=1}^M \left\| \hat{\mu}_i^{(m)} - \bar{\mu}_i \right\|^2$ ▷ EU

2.5. Feature selection

Deep learning models are generally robust to multicollinearity issues in the feature space (De Veaux and Ungar, 1994; Hamedianfar et al., 2022), thus the goal of feature selection was not to improve model performance, but (i) reducing the amount of data during training and inference, and (ii) enhancing interpretability without losing performance. Although CNNs can make manually generating complex features obsolete (Ayrey and Hayes, 2018), prior tests within this study

Table 1

Results for sequentially dropping the feature with the highest mean VIF from $n=20$ to $n=10$.

n	20	19	18	17	16	15	14	13	12	11
VIF	1771.0	523.4	348.9	218.8	147.4	147.2	147.1	105.8	75.2	61.9
	B08	MSI	SAVI	NDGI	GNDVI	EVI	RENDVI	NLI	WDRVI	B06

showed that manually engineered features such as Normalized Difference Vegetation index (NDVI) greatly improved model performance. Yet only a small set of indices, that involve a maximum of two Sentinel-2 bands, was added to the feature set. DL models generally are more computationally expensive than conventional ML algorithms, such as e.g. Random Forest or Support Vector Machines, therefore methods such as mean decrease accuracy, which involve several model runs, are not applicable. Therefore, we used the model-independent Variance Inflation Factor (VIF) for feature selection from the multivariate time series (Cheng et al., 2022). VIF was computed sequentially per time step and averaged across the time period to assess multicollinearity. We reduced the initial 20 features to a final set of features consisting of seven Sentinel-2 bands (B02, B03, B04, B05, B07, B11, B12), NDVI, Normalized Difference Moisture Index (NDMI), and Near-infrared Reflectance of Vegetation (NIRv) (Table 1; A).

2.6. Validation and uncertainty estimation

Model generalizability and robustness was tested using a spatial 5-fold cross validation within the selected protected areas. The 20 VPO training clusters were used to split training and validation while the five study areas (Fig. 4) were used to split testing datasets. Model performance was primarily measured using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) of predicted and true cover. To further investigate statistically significant differences between the models, MAE for each test dataset is used to perform a Friedman-test. With significant differences detected, a pairwise post-hoc analysis based on a Wilcoxon signed-rank test and a Holm's correction with $\alpha = 0.05$ is used to visualize critical differences between model pairs (Benavoli et al., 2016). To investigate model performance under OOD data input, a dataset with 500 samples per class were randomly sampled from a forest type map (VDB classes: mediterranean broad-leaved, submediterranean, temperate broad-leaved, temperate needle-leaved, mediterranean needle-leaved, and boreal). Visualizing the spectral-temporal feature distribution using t-distributed Stochastic Neighbor Embedding (t-SNE) (Maaten and Hinton, 2008), revealed a domain shift especially for coniferous dominated pixels, indicating OOD data (Fig. 8). We selected these plots (CON_{plots}) for evaluating the model predictions and comparing to normalized uncertainty estimates.

3. Results

3.1. Performance evaluation of predictive models across test areas

The performance of six predictive models was evaluated using RMSE and MAE of EVE cover across the five diverse test areas (Table B.5): SIB, GRA, GEN, CIL and NEB. The lowest average RMSE were achieved by $mVDB_{\text{cover}}$ (17.43%_{cover}) and $mVDB_{\text{ftype}}$ (19.08%_{cover}), demonstrating that context-related pretraining outperforms the other tested approaches. Both models consistently performed well across all regions, with the largest margins in GRA and GEN. In contrast, $mVPO2024_r$ (23.84%_{cover}) and $rVPO2024$ (23.08%_{cover}) exhibited the highest average RMSE values, indicating that non-pretrained models are less effective at capturing spatial or temporal patterns across regions. MAE results corroborate RMSE findings (Table B.4). The $mVDB_{\text{cover}}$ model yielded the lowest average MAE (12.8%_{cover}), reflecting its ability to minimize overall magnitude of prediction errors. Similarly, $mVDB_{\text{ftype}}$ achieved the second-lowest MAE (13.53%_{cover}).

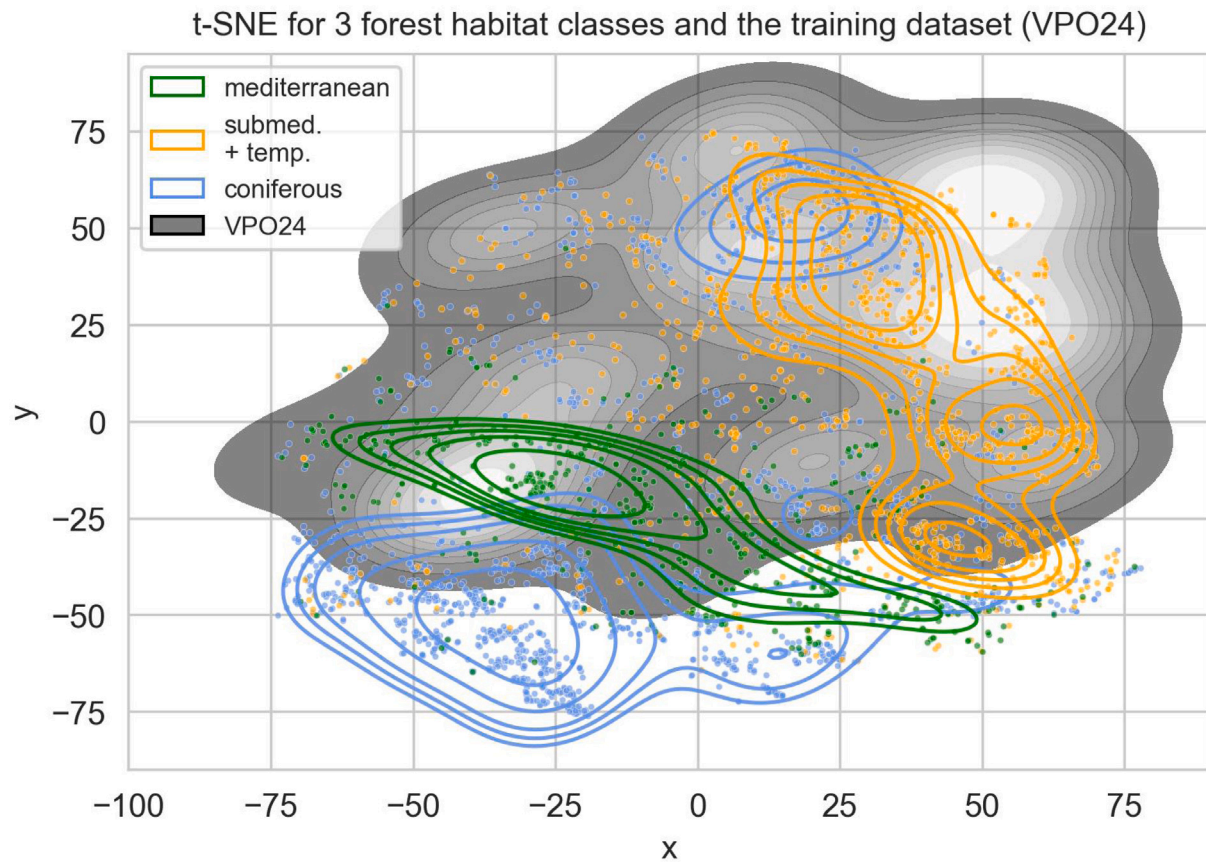


Fig. 8. t-SNE based on spectral temporal features for: sampled points from forest habitat classes, and the training dataset VPO2024. Forest habitat classes are aggregated into 3 distinct classes resembling EVE (mediterranean), deciduous (submediterranean + temperate broad-leaved), and coniferous dominated forest stands. Point kernel density is represented as contour lines to visualize overlaps of feature attributes.

Models with higher MAE values, such as $mVPO2024_r$ (18.55%_{cover}) and $rfVPO2024$ (18.32%_{cover}), underperformed. Solely looking at the tested DL models, the largest performance increase occurs when switching from random to spatial training data split, yielding an RMSE improvement ranging from 3.01–7.26%_{cover}.

Fig. 9 underlines these findings, showing that the pretrained models consistently outperform other predictors across the test regions. A Friedman test ($p=0.003$) confirmed statistically significant differences among the five model performances. A follow-up Wilcoxon signed-rank test revealed three different performance groups: ($mVDB_{cover}$ and $mVDB_{ftype}$) vs. ($mUPD$ and $mVPO2024$) vs. ($rfVPO2024$ and $mVPO2024_r$), as shown in Fig. C.19. These statistics support the hypothesis that supervised contextual pretraining holds significantly better results than the alternatively investigated methods.

In general all models overestimate EVE cover in all test sites except GEN. Error variability in SIB, GRA, and GEN is substantial and model-dependent, indicating sensitivity to model pretraining. In contrast, CIL and NEB exhibit consistently high errors across all models, suggesting site-specific prediction difficulty independent of model type, with limited improvement from pretraining (Fig. 10). Under OOD input, particularly with forest types absent from training data, but present in the test data, all models perform poorly (Fig. 11). Under Temperate broad-leaved and Mediterranean forest habitat types models predict reasonable and consistent EVE cover values, while inter-model variability for coniferous stands is high. Tested in coniferous areas, an expected EVE cover of 0%_{cover} was exceeded by all models, predicting cover values above 30%_{cover}. With $mVDB_{cover}$ and $mVDB_{ftype}$ predicting the lowest EVE cover and $mVPO2024_r$ and $mUPD$ showing the highest values. Fig. 8 indicates a feature space overlap between Mediterranean

(EVE dominated) and coniferous dominated forest areas, while parts of the coniferous input are not covered by the VPO2024 training dataset. This suggests that with OOD STS from coniferous stands entering the model the output is likely to overestimate EVE cover. To test whether this problem is mirrored in the estimated EU, EU maps were normalized for each model and test area using the 2nd and 98th percentile, and average normalized EU was computed per test area. Mediterranean and Temperate broad-leaved stands showed consistent and below average EU estimates throughout the tested models. Conversely, Coniferous samples yielded higher than average EU for all models, indicating their capability to detect error-prone regions (Fig. 12). $mVDB_{cover}$ and $mVDB_{ftype}$ estimating the highest EU, although $mVDB_{cover}$ with the highest range from average to highest values. The three other models show lower EU closer to their respective test area averages, reducing detectability of OOD-induced uncertainty during analysis of the map product. Fig. 17 underlines this statement, showing that the Common beech (*Fagus sylvatica*) dominated broad-leaved parts of the forest exhibit low EVE cover, while the Norway spruce (*Picea abies*) dominated coniferous forest patches contain substantially higher EVE cover values.

To assess whether our adapted InceptionTime model captures relevant phenological features, saliency maps were computed from gradients of the model output with respect to the input STS and visualized via average $input \times gradient$ attributions (Simonyan et al., 2014). Although varying model-dependent, the resulting patterns aligned with typical forest phenological phases (Fig. 13). NDVI shows high positive attribution during leaf-off and senescence season, indicating that the model associates high NDVI in these periods with higher EVE cover. Conversely, NDVI has negative attribution during the growing season,

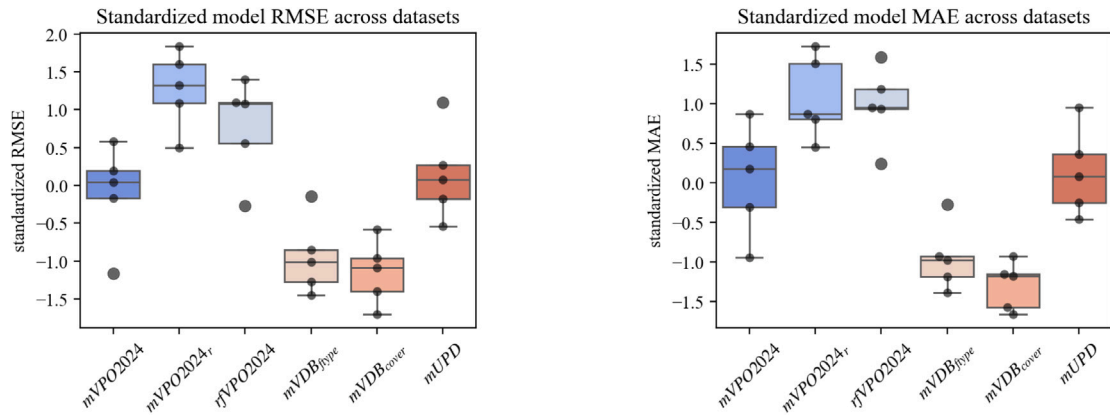


Fig. 9. Standardized model RMSE and MAE for the six different models across the five test datasets. Standardization was performed for each test dataset to help inter-comparison despite dataset performance differences.

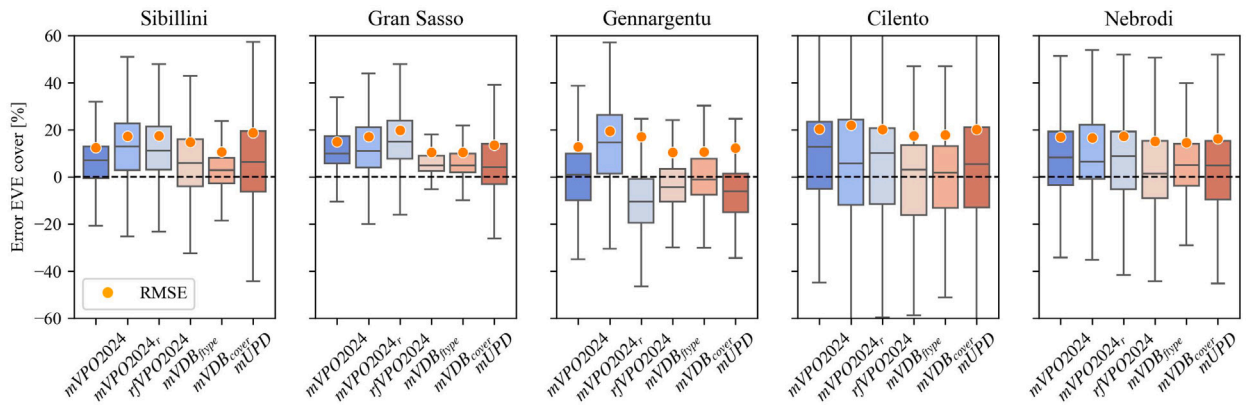


Fig. 10. EVE cover prediction error for parks (test sites) and models in the cross-validation scheme.

leading to lower EVE cover predictions. Band B05 shows an inverse pattern, with high negative attribution during leaf-off and senescence season. These attribution patterns confirm that the CIB effectively encodes phenologically relevant information for EVE cover mapping from annual multispectral STS and derived features. Comparison of gradient attributions between non-pretrained *mVPO2024* (Fig. 13A) and pretrained *mVDB_{ftype}* (Fig. 13B) shows that the latter exhibits more pronounced and temporally coherent phenological patterns. Reduced noise and clearer seasonal trends in the pretrained model indicate that its backbone has learned to extract more meaningful phenological features from the input STS.

3.2. Regional trends and model consistency

Model performance varied across regions, with *mVDB_{cover}* consistently outperforming other models in SIB, GRA and GEN, while showing slightly higher RMSE in CIL (23.4%_{cover}). The CIL region exhibited consistently higher errors across all models, likely attributable to its heterogeneous forest landscape, complex terrain, or increased variability in input data quality. Conversely, GRA and GEN displayed lower errors, suggesting homogeneous and predictable environmental characteristics that the models could capture more effectively. To test these assumptions, we conducted a correlation analysis of several environmental variables (see D). For the present dataset, the strongest Pearson correlation with RMSE was found for taxonomic diversity metrics (>0.88), followed by feature space differences (0.36–0.47) and

high cloud probability (0.31). Other variables showed weak correlations (−0.19 and 0.08). This indicates that, despite similar feature space and label distributions, the inter-plot and region-wide taxonomic diversity more strongly impact error rates than topographic or feature-space quality issues in our dataset. The performance differences across test regions show that spatial heterogeneity must be considered when interpreting model results, as regions with higher complexity and variability tend to exhibit higher prediction errors. In SIB, the pretrained *mVDB_{ftype}* underperformed compared to *mVPO2024* ($\pm 1.36\%$ _{cover} RMSE), indicating that frozen pretrained backbone weights can also decrease performance. This could either be due to poorly optimized hyperparameters or the pretrained CIB did not learn meaningful features from the pretraining dataset that could be transferred to the downstream task of predicting EVE cover. The largest variation in prediction performance across the test areas occurs in *mVPO2024* ($\pm 3.47\%$ _{cover} RMSE), while the highest consistency occurs with *rfVPO2024* ($\pm 1.36\%$ _{cover} RMSE), despite being the second worst overall model. This may result from the complex nature of hyperparameter optimization in DL models, while optimizing Random Forest models is significantly less complex, yielding higher overall error but more consistency.

3.3. Mapping product

The final mapping products reflect the diverse landscapes that can be found in the five different target areas (Fig. 14). SIB and GRA, both mountainous areas in Central Italy, show generally low EVE cover, while GEN having the highest overall cover. In CIL and NEB, high

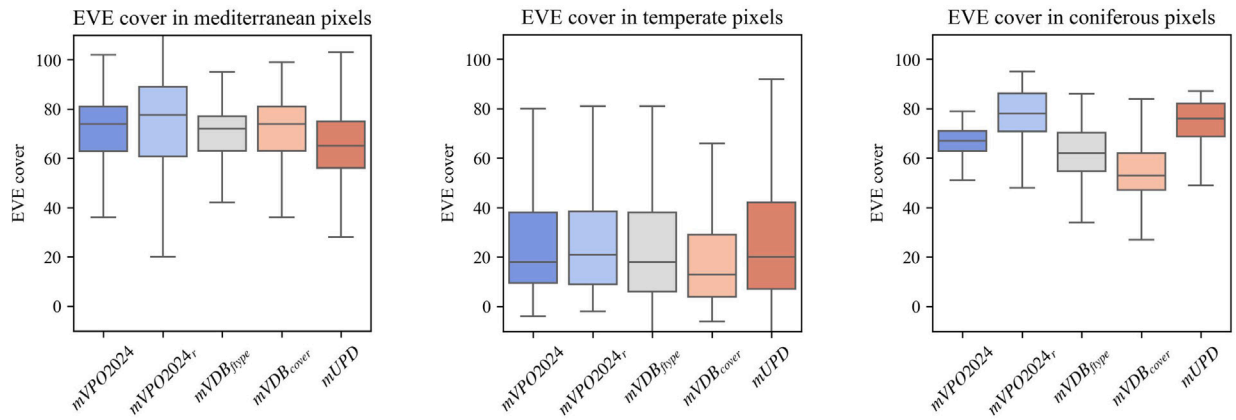


Fig. 11. EVE cover predictions for different forest habitat types.

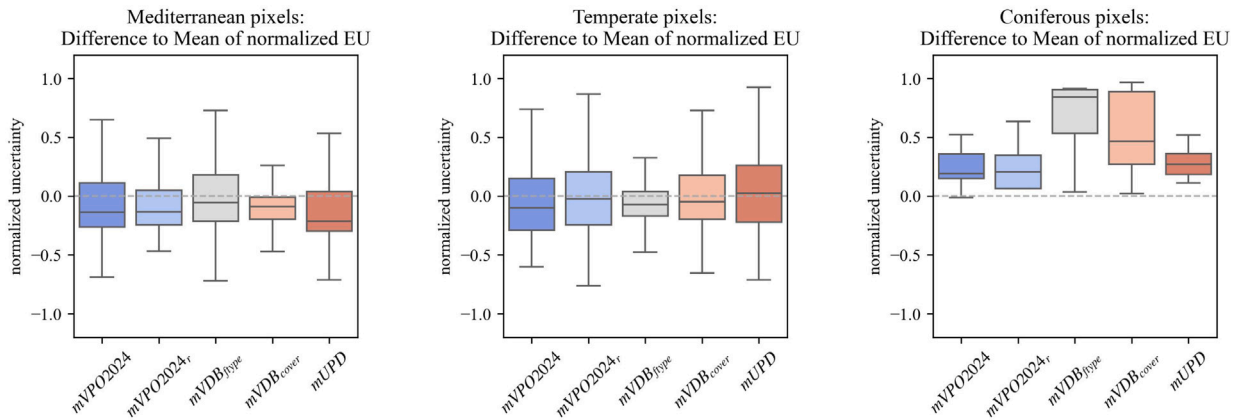


Fig. 12. Difference to Average of normalized epistemic uncertainty estimates per model type for different forest habitat types. This shows the models capability to estimate high uncertainty (above global average) for unknown forest types.

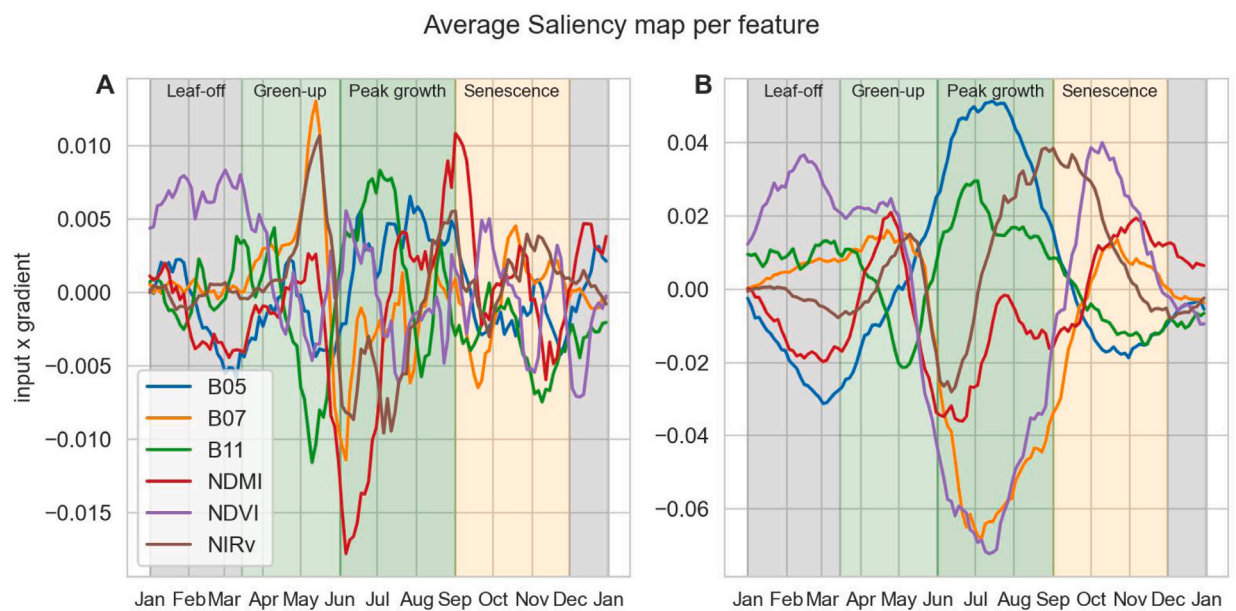


Fig. 13. Saliency maps for (A) mVPO2024 and (B) mVDBtype (testsite: CIL) and a selection of the used features. The attributions were calculated per time step and feature using $inputSTS \times gradient$ for all samples in the VPO dataset. The gradients then were averaged over all samples.

Table 2

Percentage of total valid pixel area (%; rows) covered by EVEs (%_{cover}, headings) for each model.

	0%–25%	26%–50%	51%–75%	76%–100%
<i>mV PO2024</i>	35.4	24.0	26.0	14.5
<i>mV PO2024_r</i>	34.7	26.4	20.5	18.4
<i>r fV PO2024</i>	36.3	29.2	29.6	4.9
<i>mV DB_{ftype}</i>	37.8	24.8	26.1	11.3
<i>mV DB_{cover}</i>	43.9	22.6	18.9	14.6
<i>mU PD</i>	35.9	25.9	27.8	10.3
	37.3	25.5	24.8	12.3

EVE cover is observed in the lower elevated, coastal regions, while EVE cover decreases with elevation and distance to the Mediterranean sea. Table 2 shows that approximately one-third of all mapped pixels have EVE cover lower than 25%_{cover}, while 40% to 60% have cover values ranging between 26%_{cover} and 75%_{cover}, depending on the model. Despite the high overall EVE cover in GEN, only between 4.9% to 15.4% of its mapped area exceeds 75%_{cover}. Discrepancies between spatial distribution of model predictions increases with EVE cover. Likely this is an effect related to the underrepresentation of high EVE cover plots in the training datasets cover value distribution. In this exploratory study on pretraining impact, no additional augmentation step was taken to address imbalance or underrepresentation, although other research has shown that techniques like oversampling and augmenting underrepresented parts of data can improve overall prediction performance (Lang et al., 2023; Safonova et al., 2023). Fig. 15 shows the standard deviation (SD) across model predictions for the western part of NEB. In all parks, high inter-model variability concentrates in forest-nonforest transition zones and mixed forest areas, indicating that the difference between the models is spatially consistent. Taking all test areas into account, in 92% of valid pixels inter-model agreement lies within a SD of 0–15%_{cover}, but only 22% of pixels exhibit a SD lower than 5%_{cover}. Both pixel-wise SD and spatial distribution of predictions underscore the substantial divergence among models trained on the same dataset. This reflects a broader problem for models trained on plot observations sampled over large areas, as a sufficient coverage of the areas target characteristics and corresponding spectral variability is difficult to achieve. While differently initialized or parametrized models generally agree in areas with high data coverage, the predictions diverge in insufficiently covered areas.

3.4. Uncertainty estimates

Regions with high EU estimates can be divided into three categories (Fig. 16): (i) non-forested areas, (ii) areas with obscured reflectance, and (iii) underrepresented forest types. Although a forest mask was applied prior to model deployment, mapping non-forested pixels could not be prevented. Prediction results for such pixels are error-prone to useless. A second low-confidence area are steep north-facing slopes with obscured spectral reflectance time series, likely due to a lack of spectral information during winter and resulting gaps in phenological information. The third high uncertainty area includes underrepresented forest characteristics in the training data, which involves varying leaf-morphology types or mixed forest stands. In these cases, uncertainty estimates effectively capture the insufficient coverage of the training dataset. In contrast, AU, primarily capturing labeling noise and ambiguity, shows a different spatial pattern. Spatial correlation analysis between AU and Sentinel-2 L2A SCL frequencies reveals no consistent link to typical sources of data noise (e.g. high cloud frequency, shadow, or dark pixels). Class-wise comparisons between different forest habitat types shows that Temperate deciduous forests exhibit the lowest AU estimates, likely due to their phenologically distinct spectral-temporal signatures and higher label certainty (value and location-wise). In contrast, submediterranean and coniferous stands tend to have higher

AU estimates (Fig. 18). In mixed deciduous and evergreen submediterranean forests the connection between labeled EVE cover value and the extracted spectral-temporal features is less accurate, due to their heterogeneous nature, where small in-plot location measurement errors can have high impact on spectral-temporal features.

While AU calibration relies on training loss, EU arises from ensemble diversity and is sensitive to training hyperparameters such as dropout rates or training time, which are optimized for predictive performance and AU, but not explicitly EU. As this type of uncertainty mainly refers to data that is not contained in the training dataset, an evaluation or post-hoc calibration of values is hard to achieve in the absence of additional reference data (Yang and Yee, 2024). In general, trained models are overconfident for predictions with large true error ($> 10\%_{cover}$) and underconfident for predictions with true error $< 10\%_{cover}$.

4. Discussion

4.1. Uncertainty estimation and its significance

During the training process, dropout, random initialization of model heads and independent shuffling of batches (Lakshminarayanan et al., 2017) play a crucial role in maintaining model diversity in EU estimation in ensemble-based approaches. This diversity enhances interpretability of results in transition zones and areas with OOD characteristics, where manual thresholding may be necessary. While EU provides spatially meaningful insights, reliable calibration is difficult and correlation to actual prediction errors is weak (Yang and Yee, 2024). In contrast, AU captures label noise, but remains challenging to interpret, as it often does not follow a meaningful spatial pattern. Complementing findings from other research fields (Charpentier et al., 2022; Mucsányi et al., 2025; Valdenegro-Toro and Mori, 2022), separating EU and AU can help interpretation, given the difficulty of reliable uncertainty calibration and its scaling issues. If not disentangled, uncertainty estimates may conflate different sources of error, complicating interpretation. This distinction is particularly relevant when considering the trade-off between model calibration and uncertainty estimation as two separate but complementary aspects of performance evaluation. In a sparse data context, calibrating uncertainty to training/test error may neglect the primary goal of identifying OOD areas or regions with unreliable input data, as calibrating EU on test data error may risk ignoring natural uncertainty estimates coming from inherent data variability. For biogeographic applications, the choice of uncertainty estimation should align with the goal of capturing meaningful ecological patterns/transitions and inconsistencies in the final map product. Therefore relative calibration might be more important than absolute calibration, as manual thresholding is necessary regardless, if unreliable predictions have to be discarded. Sylvain et al. (2024) referred to their ensemble uncertainty as level of ensemble agreement, which reflects the reliability of the ensembles prediction. In our particular use case the approach proves useful in detecting OOD time series and spatial inconsistencies. A reliable uncertainty estimation that correlates to actual prediction errors not only in a relative manner is challenging to calibrate during training but also in post processing (Yang and Yee, 2024). This rises the question if in a non-sensitive application like mapping of vegetation properties, relative uncertainty is sufficient. A comprehensive point dataset capturing all characteristics and variability of a forest at all scales is unattainable. Therefore model predictions naturally carry uncertainty, spatially explicit quantification of this uncertainty gives us the possibility to focus interpretation of mapped model results to reliable predictions. The EU estimates generated in this study effectively highlight the model's confidence across different landscapes. High-uncertainty regions often correspond to areas with a lack of knowledge about the input data time series. This could either refer to corrupted spectral-temporal signals (e.g. with bad quality data in shaded areas) or a spectral-temporal signal that originates from a surface that is

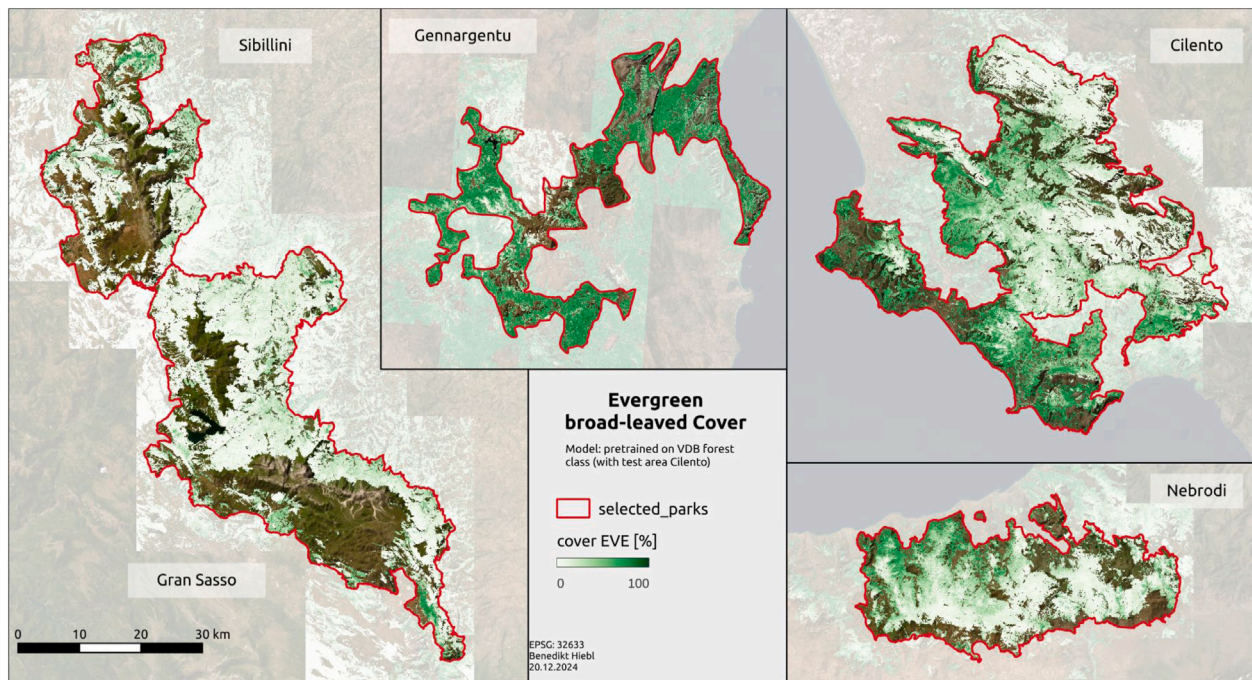


Fig. 14. EVE cover mapped to the five target areas using $mVDB_{f_{type}}$.

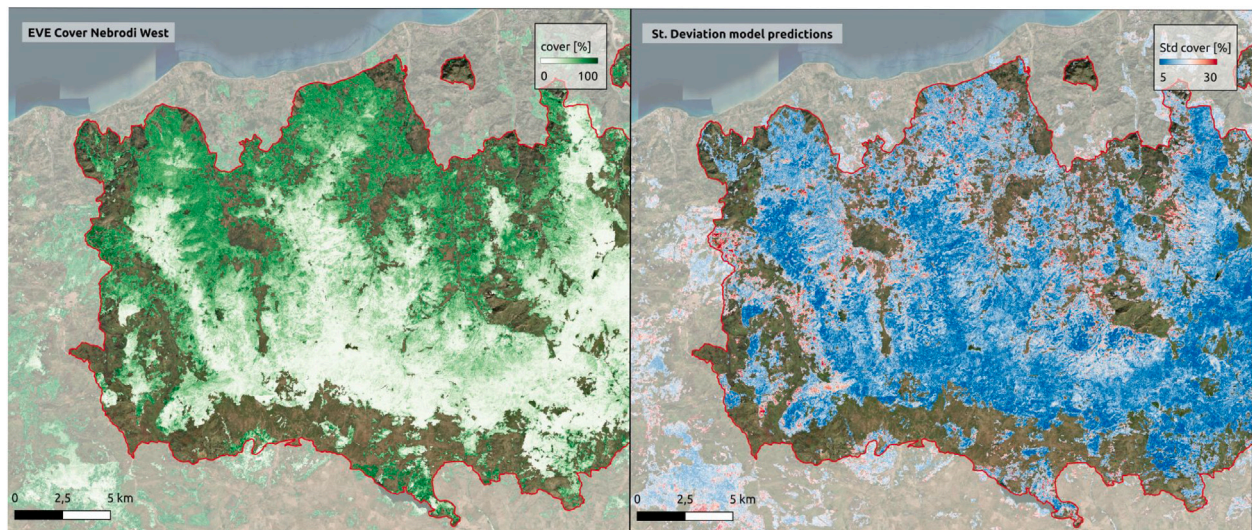


Fig. 15. Standard deviation across all model predictions for the western part of NEB.

not contained in the training data (here e.g. non-forested surfaces, or not sampled forest habitats). These findings suggest that EU can guide targeted data acquisition, improving model performance in challenging environments. Integrating EU-driven data acquisition into the sampling design and data collection process for ML applications ensures that regions with high uncertainty are prioritized for additional ground-truth annotation. By focusing data collection efforts on uncertain areas, knowledge gaps in the training set can be systematically reduced and enhance the model's ability to generalize. Developing a sequential uncertainty-driven sampling approach for mapping in distinct areas, could reduce data needs and therefore labor intensive field work that is often a major limitation in environmental mapping. This approach aligns with Active Learning methods (Tuia et al., 2011), where the

model itself actively selects the most informative samples. Additionally, combining AU and EU can help distinction between persistent uncertainty due to inherent class variability and reducible uncertainty caused by limited training data. Evaluating the impact of targeted data collection through an iterative training process would provide a structured way to quantify improvements in model confidence and prediction metrics. By utilizing EU estimates for guiding data collection, mapping studies can develop more efficient, adaptive, and probabilistic deep learning models for RS applications, ultimately enhancing their reliability and reducing the burden of extensive field sampling. For practitioners, absolute AU rather than the relative spatial distribution of AU of a mapped product gives information on the models error based on the bias that comes with the specific characteristics of the

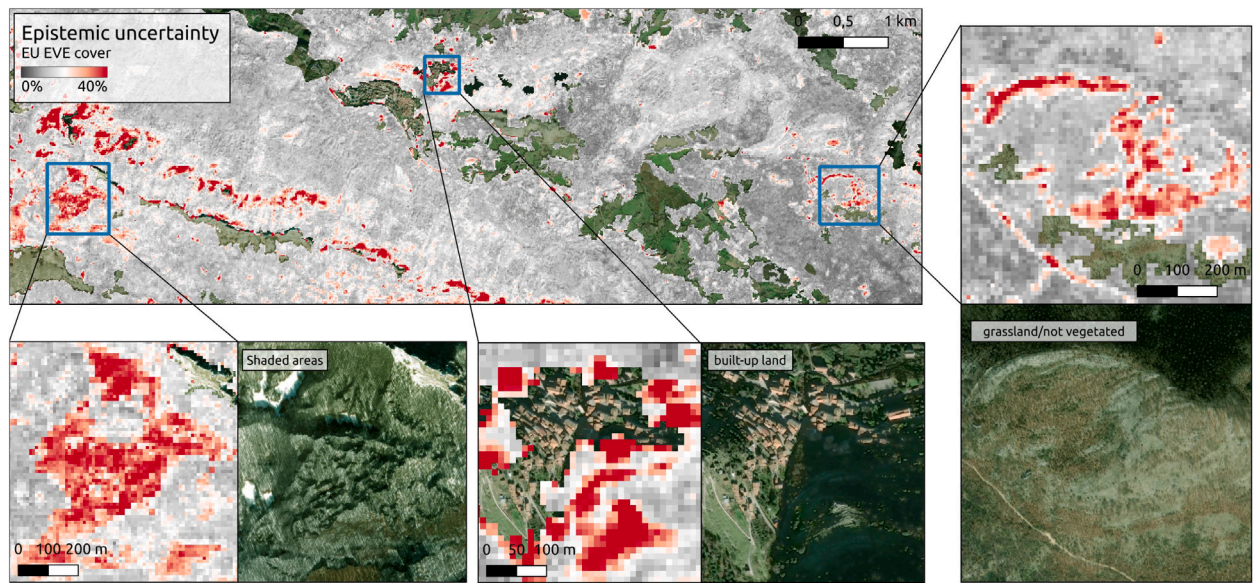


Fig. 16. Different types of surface signals that correspond to high epistemic uncertainty in CIL.

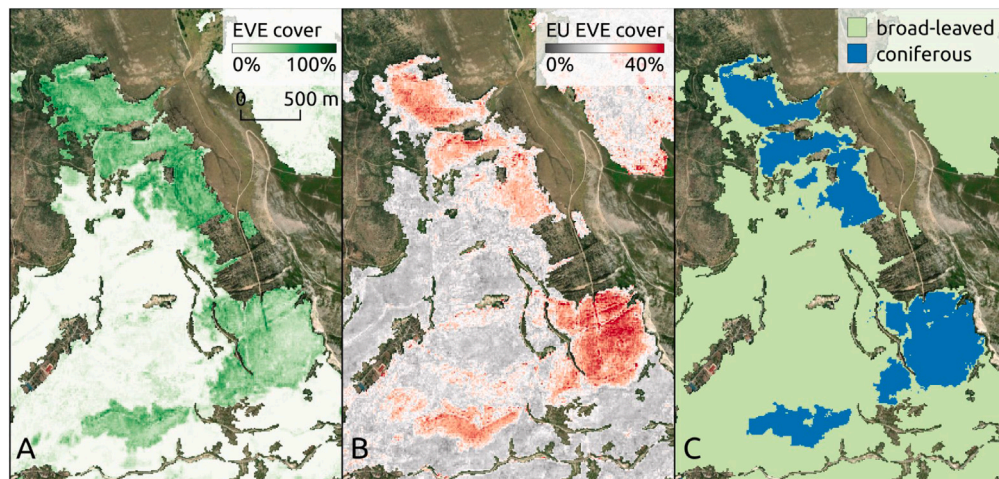


Fig. 17. Prediction results in coniferous forest patches are generally wrong in all models. A and B showing EVE cover and EU for $mVDB_{type}$ in a temperate forest in SIB. C outlines the areas with coniferous dominated forest based on COPERNICUS forest type map.

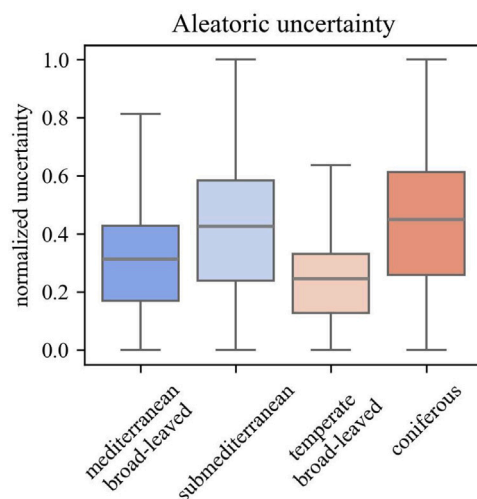


Fig. 18. Normalized aleatoric uncertainty estimates for the test areas and $mVDB_{type}$.

sampled areas (e.g. heterogeneity of the vegetation) or the labeling accuracy of the plot observations. In contrast, spatial distribution of EU estimates provide meaningful insights into problematic prediction areas (e.g. corrupted data input), data gaps and can have a large impact on product analysis when accounted for.

4.2. Performance in OOD regions

A major challenge in RS-based species mapping is model generalization to OOD data, where environmental and spectral conditions differ from those represented in the training dataset (Gao et al., 2024; Ma et al., 2024). Our results confirm that models struggle in pure coniferous stands where EVEs are absent, with predicted cover values substantially overestimated. One reason for including uncertainty estimates and probabilistic instead of deterministic modeling was to assess their utility to detect OOD data input, particularly in forest types not present in the VPO2024 dataset. In this context, two questions arise: (a) Does the chosen pretraining strategies improve performance in these areas? (b) are uncertainty estimates capturing the high error in these areas? Referring to Fig. 12, all model types perform poorly in coniferous stands, where EVE cover should trend towards 0%_{cover}. Two pretrained

models tend to predict lower EVE cover values, but the difference is insufficient to claim clear improvement. Fig. 17 supports this statement, while the Common beech (*Fagus sylvatica*) dominated broad-leaved parts of the forest exhibit low EVE cover, the Norway spruce (*Picea abies*) dominated coniferous forest patches contain erroneously substantially higher EVE cover values. These findings show, that ML models trained on plot observations propagate dataset gaps to the final map product, potentially corrupting analysis, although the present case is an extreme example. In line with Guo and Yu (2022), it can be concluded that the method chosen for pretraining alone, while improving spatial generalizability, did not significantly improve overall performance on the selected OOD data. A pixel-wise uncertainty estimate can highlight these error-prone OOD regions, if detected reliably. The final EVE cover maps and corresponding uncertainty estimates indicate, that EU in coniferous forest patches can be used to detect these unreliable predictions, enabling exclusion through thresholding. However, this approach also excludes other high-EU areas, reducing usable coverage of the map product. The simple method chosen to estimate EU via an ensemble of model heads is capable of capturing these problems via relative uncertainty, though proper calibration remains challenging. A post-hoc calibration as, for example, proposed by Yang and Yee (2024) may in many cases be impossible as an additional dataset that contains parts of the OOD data is required. Manual adjustment or interpretation of EU maps in combination with the target variable maps (here: EVE cover) is necessary. More sophisticated methods such as Bayesian Neural Networks could improve calibration (Deshpande et al., 2024; Ma et al., 2021), but involve a deeper understanding of DL and Bayesian methods.

4.3. Effects of spatial splitting and pretraining task alignment

The largest performance improvement was observed when applying a spatial split of the training data based on longitude/latitude clusters, indicating that this approach mitigates overfitting caused by spatial autocorrelation. This underscores the critical role of hyperparameter optimization in training the model independent of pretraining, as performance discrepancies between validation and test data are higher with random splits. While fair training and testing splits have been suggested in other studies (Salazar et al., 2022), the present study is not constructed around a fair testing site, as the sparse availability of training data imposes an unfair deployment of the model in areas with an inevitable domain shift (Ma et al., 2024). However, the trade-off between preserving information and preventing model overfitting should be further explored in different environmental settings. But we found that, model selection and optimizing hyperparameters aiming for a better model generalization can be improved with this approach. By splitting the dataset based on random spatial clusters, the specific characteristics of distinct areas that fall into the validation dataset may be lost. Careful adjustment is required to avoid imbalances in the training and validation dataset. RMSE and MAE showed substantial variance between test regions, even for the same model configuration, indicating sensitivity to cluster choice and size, although improving overall generalization capabilities. MAE and RMSE results confirm that the closer the pretraining task aligns with the target task, the better the model transfers knowledge, which is consistent with prior work (Ma et al., 2024; Yosinski et al., 2014). This finding is promising, as large datasets featuring a variety of environmental and vegetation variables are freely available (Agrillo et al., 2017; Jandt et al., 2022; Sabatini et al., 2021) and could be leveraged for pretraining to enhance large-scale environmental and vegetation mapping. This could be implemented on a much larger scale than implemented in this study, combining different datasets across studies. Although this assumption is not statistically significant, due to the small test setting of the present study. In our case, the unsupervised method (mUPD) could not learn transferable features that could be used directly in the target task. This problem could be caused by the chosen fine-tuning method. According

to Yosinski et al. (2014), later layers in the model are more target specific while the first layers learn more general features. By freezing different layers or fine-tuning the whole model instead of only the heads, mUPD might also benefit from the unsupervised pretraining.

4.4. Challenges in generalization and model transferability

The substantial performance variations across different test areas highlights the importance of well-chosen training, validation, and test splits. The observed differences between parks suggest that, despite the apparent simplicity of the task and the assumption of spatial invariance, the models fail to generalize effectively when specific parks are excluded from training. This indicates a lack of critical contextual information that is spatially variable between regions. While this is present in all tested models, context-related pretraining ($mVDB_{ftype}$ and $mVDB_{cover}$) shows that the information contained in the larger VDB database can be transferred and preserved to improve generalization capability to areas that are outside the training data scope. Pretraining on a larger, more diverse dataset mitigates a covariate domain shift (Ma et al., 2024) between training and test data, despite the presence of a covariate, prior and concept domain shift from the pretrained model to the fine-tuned model. However, the analysis of error sources for cross-park RMSE showed that domain shifts in feature space and especially vegetation diversity introduce a performance degradation that cannot be compensated by our pretraining setup. The potential for foundation models in this context remains an open question. Although they offer advantages for broad applications, adapting them to specific sensor and feature characteristics remains challenging. Recent work on flexible Transformer architectures, such as EarthPT or Presto Smith et al. (2024), Tseng et al. (2024), aim to address this issue, but further research is necessary to evaluate their effectiveness in RS vegetation analysis. Specific RS tasks require custom models, sensors and features, where flexibility of foundation models may be limited. Our results suggest that pretraining custom models on datasets, which are contextually related to the specific target task, is a practical and flexible way to improve task-specific model performance in RS applications.

5. Conclusion

This study demonstrates the effectiveness of pretraining strategies for deep learning-based mapping of EVE tree and shrub species cover in mixed forests. Models pretrained on a diverse forest vegetation database significantly outperform those trained solely on small in-situ datasets, particularly in terms of spatial generalization to new regions. Moreover, our results show the utility of EU estimates for identifying areas with OOD predictions, improving the interpretability and reliability of large-scale mapping products. However, challenges remain in optimizing spatial generalization across heterogeneous forest types and handling OOD data input. Future work may explore the potential of large-scale foundation models and adaptive, uncertainty-driven sampling strategies to further improve predictive performance and uncertainty estimation in RS-based vegetation monitoring, where field work data acquisition is a major limitation. Further, automating the disentanglement of uncertainty estimates into interpretable thematic patterns, such as “topographic influences” or “uncertainties from incomplete observations”, could enhance their relevance in decision-sensitive applications. Understanding the sources of uncertainty can reveal underlying dataset biases, guide model improvements, and inform risk-aware decision-making in environmental monitoring and resource management. To address spatial autocorrelation, we used constrained k-means clustering. While effective in our scenario, further research is needed to assess the impact of minimum distance thresholds between clusters and the relation between sampling size and cluster size.

CRediT authorship contribution statement

Benedikt Hiebl: Writing – original draft, Visualization, Software, Methodology, Conceptualization. **Nicola Alessi:** Writing – review & editing, Data curation. **Giacomo Calvia:** Writing – review & editing, Data curation. **Alessandro Bricca:** Writing – review & editing. **Gianmaria Bonari:** Writing – review & editing. **Giulio Zangari:** Writing – review & editing. **Wouter Dorigo:** Writing – review & editing. **Stefan Zerbe:** Writing – review & editing, Project administration, Funding acquisition. **Martin Rutzinger:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used OpenAIs ChatGPT and DeepL to improve sentence readability, translate words and sentences, and find synonyms for often used phrases. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research has been conducted within the project “TRACEVE - Tracing the evergreen broad-leaved species and their spread” (I 6452-B) funded by the Austrian Science Fund (FWF). GB was funded under the National Recovery and Resilience Plan (NRRP), Mission 4 Component 2 Investment 1.4—Call for tender No. 3138 of 16 December 2021, rectified by Decree n. 3175 of 18 December 2021 of Italian Ministry of University and Research funded by the European Union – NextGenerationEU. Project code CN_00000033, Concession Decree No. 1034 of 17 June 2022 adopted by the Italian Ministry of University and Research, CUP B63C22000650007, Project title “National Biodiversity Future Center—NBFC”.

The computational results presented here have been achieved (in part) using the LEO5 HPC infrastructure of the University of Innsbruck.

Appendix A. Vegetation indices

Additionally to 9 Sentinel-2 bands (B02, B03, B04, B05, B06, B07, B08, B11, B12), 11 vegetation indices were considered and calculated for the initial set of features (Montero et al., 2023; Yang et al., 2019;

Huete et al., 1997; Badgley et al., 2017; Camps-Valls et al., 2021; Gitelson, 2004; Wilson and Sader, 2002) (Table A.3).

Appendix B. MAE and RMSE of model predictions

See Tables B.4 and B.5

Appendix C. Critical difference of model performances

See Fig. C.19.

Appendix D. Dominant sources of error

To investigate across-park dominant error sources, we analyzed several variables related to topography, STS input quality, test to training data feature domain shift, and vegetation heterogeneity regarding there correlation to per park RMSE. We retrieved following variables plot-wise per test site:

- average slope (*Slope*)
- average Terrain Ruggedness Index (*TRI*)
- Sentinel-2 L2A SCL class “dark pixel” frequency ($dark_{scl}$)
- Sentinel-2 L2A SCL class “medium cloud probability” frequency (mc_{scl})
- Sentinel-2 L2A SCL class “high cloud probability” frequency (hc_{scl})
- mean pairwise euclidean distance between test and training + validation dataset after Principal Component (PCA) decomposition of spectral–temporal features (PCA_{all})
- mean pairwise euclidean distance between test and validation dataset after PCA decomposition of spectral–temporal features (PCA_{val})
- Inter-plot taxonomic β -diversity based on species cover (Jensen–Shannon Diversion) (βDiv)
- Park-wide taxonomic γ -diversity (Shannon-Index) (γDiv)

As shown in Table D.6, the highest Pearson correlation to RMSE was found for the taxonomic diversity metrics of γDiv and βDiv (0.88 and 0.91), followed by feature space differences (0.35 and 0.47). For STS input data quality only high cloud probability (0.36) exhibited a weak correlation, while dark pixels, and medium cloud probability had no correlation to RMSE in this dataset. Topographic factors of average slope and average Terrain Ruggedness Index showed very low negative correlation. These findings should be taken with care as there is no evidence that the statistics of the dataset in this study can be transferred to other datasets.

Table A.3
Initially considered vegetation indices before feature selection.

Name	Description	Sentinel-2 Bands	Formula
NDVI	Normalized Difference Vegetation Index	B04, B08	$\frac{B08 - B04}{B08 + B04}$
NDMI	Normalized Difference Moisture Index	B08, B11	$\frac{B08 - B11}{B08 + B11}$
NIRv	Near-Infrared Reflectance of Vegetation	B04, B08	$NDVI \times B08$
MSI	Moisture Stress Index	B08, B11	$\frac{B11}{B08}$
SAVI	Soil-Adjusted Vegetation Index	B04, B08	$\frac{(1+L)(B08-B04)}{B08+B04+L}$
NDGI	Normalized Difference Greenness Index	B03, B04, B08	$\frac{\epsilon B03 + (1-\epsilon)B08 - B04}{\epsilon B03 + (1-\epsilon)B08 + B04}$
GNDVI	Green NDVI	B03, B08	$\frac{B08 - B03}{B08 + B03}$
EVI	Enhanced Vegetation Index	B02, B04, B08	$\frac{\epsilon(B08 - B04)}{B08 + C_1 B04 - C_2 B02 + L}$
RENDVI	Red Edge NDVI	B05, B06	$\frac{B06 - B05}{B06 + B05}$
NLI	Non-linear Vegetation Index	B04, B08	$\frac{B08^2 - B04}{B08^2 + B04}$
WDRVI	Wide Dynamic Range Vegetation Index	B04, B08	$\frac{\alpha B08 - B04}{\alpha B08 + B04}$

Table B.4
MAE (in %_{cover}) and their model and area dependent standard deviation (in %_{cover}) over all model types and test areas. Bold indicating the best model.

	SIB	GRA	GEN	CIL	NEB	Mean	Rank
<i>mV PO2024</i>	12.33	14.99	12.72	20.35	16.92	15.46 ± 3.30	3
<i>mV PO2024_r</i>	17.31	17.12	19.84	21.94	16.52	18.55 ± 2.28	6
<i>rfV PO2024</i>	17.54	19.72	17.09	20.01	17.23	18.32 ± 1.43	5
<i>mV DB_ftype</i>	14.18	10.38	10.40	17.52	15.18	13.53 ± 3.11	2
<i>mV DB_{cover}</i>	10.59	10.51	10.58	17.85	14.48	12.80 ± 3.29	1
<i>mU PD</i>	18.71	13.54	12.22	20.20	16.16	16.33 ± 2.68	4
Mean	14.92	14.38	13.81	19.65	16.08	–	–

Table B.5
RMSE (in %_{cover}) and their model and area dependent standard deviation (in %_{cover}) over all model types and test areas. Bold indicating the best model.

	SIB	GRA	GEN	CIL	NEB	Mean	Rank
<i>mV PO2024</i>	15.96	19.87	17.23	24.80	21.17	19.81 ± 3.47	3
<i>mV PO2024_r</i>	21.36	23.52	24.49	27.81	22.00	23.84 ± 2.54	6
<i>rfV PO2024</i>	23.32	24.60	22.37	23.96	21.15	23.08 ± 1.36	5
<i>mV DB_ftype</i>	19.30	15.33	13.77	21.80	19.56	17.95 ± 3.30	2
<i>mV DB_{cover}</i>	15.19	15.93	14.01	23.40	18.61	17.43 ± 3.74	1
<i>mU PD</i>	25.34	19.10	15.71	24.94	20.61	21.27 ± 2.79	4
Mean	19.74	19.73	17.93	24.45	20.52	–	–

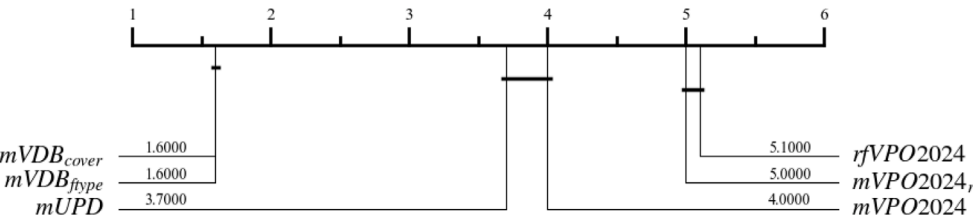


Fig. C.19. Critical Difference Diagram based on MAE for the test datasets. Low ranking indicates better performance. Thick lines indicate groups of models with no statistically significant performance difference.

Table D.6
Per plot possible error sources and their Pearson correlation to testsite RMSE.

	RMSE	<i>Slope</i>	<i>TRI</i>	<i>dark_{scl}</i>	<i>mcl_{scl}</i>	<i>hcl_{scl}</i>	<i>PCA_{all}</i>	<i>PCA_{val}</i>	<i>βDiv</i>	<i>γDiv</i>
SIB	19.74	27	13	0.06	0.03	0.06	7.9	8.0	0.68	17.5
GRA	19.73	21	9	0.03	0.04	0.07	7.2	7.4	0.69	12.4
GEN	17.93	22	10	0.01	0.04	0.06	6.4	6.9	0.61	6.8
CIL	24.45	20	9	0.02	0.04	0.07	7.3	7.7	0.74	24.6
NEB	20.52	16	7	0.002	0.04	0.10	6.9	7.3	0.69	10.6
Pearson		−0.19	−0.16	−0.08	0.08	0.31	0.36	0.47	0.91	0.88

Data availability

Parts of the mapping results for EVE cover including the combined estimates of AU and EU for the present study can be explored interactively under <https://benedikthiebl.users.earthengine.app/view/test>. Python code including model architecture, training agents and pre-processing steps are available under https://git.uibk.ac.at/c7161037/trave_pretraining. Training data can be downloaded from <https://doi.org/10.5281/zenodo.15000771>.

References

Abdalla, A., Cen, H., Wan, L., Rashid, R., Weng, H., Zhou, W., He, Y., 2019. Fine-tuning convolutional neural network with transfer learning for semantic segmentation of ground-level oilseed rape images in a field with high weed pressure. *Comput. Electron. Agric.* 167, 105091. <http://dx.doi.org/10.1016/j.compag.2019.105091>, URL: <https://www.sciencedirect.com/science/article/pii/S0168169919317107>.
Adagbasa, E.G., Adelabu, S.A., Okello, T.W., 2022. Application of deep learning with stratified K-fold for vegetation species discrimination in a protected mountainous region using sentinel-2 image. *Geocarto Int.* 37 (1), 142–162. <http://dx.doi.org/10.1080/10106049.2019.1704070>, Publisher: Taylor & Francis _eprint.
Agrillo, E., Alessi, N., Massimi, M., Spada, F., Sanctis, M.D., 2017. Nationwide vegetation plot database – sapienza university of rome: state of the art, basic figures

and future perspectives. *Phytocoenologia* 47 (2), 221, URL: https://www.academia.edu/35299044/Nationwide_Vegetation_Plot_Database_Sapienza_University_of_Rome_state_of_the_art_basic_figures_and_future_perspectives_Long_Database_Report.
Alessi, N., Bonari, G., Zannini, P., Jiménez-Alfaro, B., Agrillo, E., Attorre, F., Canullo, R., Casella, L., Cervellini, M., Chelli, S., et al., 2023. Probabilistic and preferential sampling approaches offer integrated perspectives of Italian forest diversity. *J. Veg. Sci.* 34 (1), e13175, Publisher: Wiley Online Library.
Alessi, N., Těšitel, J., Zerbe, S., Spada, F., Agrillo, E., Wellstein, C., 2019. Ancient refugia and present-day habitat suitability of native laurophylls in Italy. *J. Veg. Sci.* 30 (3), 564–574. <http://dx.doi.org/10.1111/jvs.12743>, _eprint: URL: <https://onlinelibrary.wiley.com/doi/abs/10.1111/jvs.12743>.
Atkinson, P.M., Jeganathan, C., Dash, J., Atzberger, C., 2012. Inter-comparison of four models for smoothing satellite sensor time-series data to estimate vegetation phenology. *Remote Sens. Environ.* 123, 400–417. <http://dx.doi.org/10.1016/j.rse.2012.04.001>, URL: <https://www.sciencedirect.com/science/article/pii/S0034425712001629>.
Ayrey, E., Hayes, D.J., 2018. The use of three-dimensional convolutional neural networks to interpret lidar for forest inventory. *Remote. Sens.* 10 (4), 649, Publisher: MDPI.
Babst, F., Bouriaud, O., Poulter, B., Trouet, V., Girardin, M.P., Frank, D.C., 2019. Twentieth century redistribution in climatic drivers of global tree growth. *Sci. Adv.* 5 (1), eaat4313. <http://dx.doi.org/10.1126/sciadv.aat4313>, Publisher: American Association for the Advancement of Science URL: <https://www.science.org/doi/full/10.1126/sciadv.aat4313>.
Badgley, G., Field, C.B., Berry, J.A., 2017. Canopy near-infrared reflectance and terrestrial photosynthesis. *Sci. Adv.* 3 (3), e1602244, Publisher: American Association for the Advancement of Science.

- Bae, S., Levick, S.R., Heidrich, L., Magdon, P., Leutner, B.F., Wöllauer, S., Serebryanyk, A., Nauss, T., Krzystek, P., Gossner, M.M., Schall, P., Heibl, C., Bässler, C., Doerfler, I., Schulze, E.-D., Krah, F.-S., Culmsee, H., Jung, K., Heurich, M., Fischer, M., Seibold, S., Thorn, S., Gerlach, T., Hothorn, T., Weisser, W.W., Müller, J., 2019. Radar vision in the mapping of forest biodiversity from space. *Nat Commun* 10 (1), 4757. <http://dx.doi.org/10.1038/s41467-019-12737-x>, Number: 1 Publisher: Nature Publishing Group URL: <https://www.nature.com/articles/s41467-019-12737-x>.
- Benavoli, A., Corani, G., Mangili, F., 2016. Should we really use post-hoc tests based on mean-ranks? *J. Mach. Learn. Res.* 17 (5), 1–10, URL: <http://jmlr.org/papers/v17/benavoli16a.html>.
- Bernico, M., Li, Y., Zhang, D., 2019. Investigating the impact of data volume and domain similarity on transfer learning applications. In: Arai, K., Bhatia, R., Kapoor, S. (Eds.), *Proceedings of the Future Technologies Conference 2018*, FTC, Springer International Publishing, Cham, pp. 53–62. http://dx.doi.org/10.1007/978-3-030-02683-7_5.
- Bolyn, C., Lejeune, P., Michez, A., Latte, N., 2022. Mapping tree species proportions from satellite imagery using spectral-spatial deep learning. *Remote Sens. Environ.* 280, 113205. <http://dx.doi.org/10.1016/j.rse.2022.113205>, URL: <https://linkinghub.elsevier.com/retrieve/pii/S0034425722003145>.
- Bradley, P.S., Bennett, K.P., Demiriz, A., 2000. Constrained K- Means Clustering. Microsoft research Technical Report, URL: <http://machinelearning102.pbworks.com/f/ConstrainedKMeanstr-2000-65.pdf>.
- Camps-Valls, G., Campos-Taberner, M., Moreno-Martínez, A., Walther, S., Duveiller, G., Cescatti, A., Mahecha, M.D., Muñoz-Marí, J., García-Haro, F.J., Günter, L., et al., 2021. A unified vegetation index for quantifying the terrestrial biosphere. *Sci. Adv.* 7 (9), eabc7447, Publisher: American Association for the Advancement of Science.
- Chabalala, Y., Adam, E., Kganayo, M., 2023. Mapping fruit tree dynamics using phenological metrics from optimal sentinel-2 data and deep neural network. *CABI Agric. Biosci.* 4 (1), 51. <http://dx.doi.org/10.1186/s43170-023-00193-z>.
- Charpentier, B., Senanayake, R., Kochenderfer, M., Günnemann, S., 2022. Disentangling epistemic and aleatoric uncertainty in reinforcement learning. <http://dx.doi.org/10.48550/arXiv.2206.01558>, URL: <http://arxiv.org/abs/2206.01558>, arXiv:2206.01558 [cs].
- Cheng, J., Sun, J., Yao, K., Xu, M., Cao, Y., 2022. A variable selection method based on mutual information and variance inflation factor. *Spectrochim. Acta Part A: Mol. Biomol. Spectrosc.* 268, 120652. <http://dx.doi.org/10.1016/j.saa.2021.120652>, URL: <https://www.sciencedirect.com/science/article/pii/S1386142521012294>.
- Chytrý, M., Tichý, L., Hennekens, S.M., Knollová, I., Janssen, J.A., Rodwell, J.S., Peterka, T., Marcondè, C., Landucci, F., Danihelka, J., et al., 2020. EUNIS habitat classification: Expert system, characteristic species combinations and distribution maps of European habitats. *Appl. Veg. Sci.* 23 (4), 648–675, Publisher: Wiley Online Library.
- Conedera, M., Colombaroli, D., Tinner, W., Krebs, P., Whitlock, C., 2017. Insights about past forest dynamics as a tool for present and future forest management in Switzerland. *Ecology of Mountain Forest Ecosystems in Europe*, Forest Ecol. Manag. Ecology of Mountain Forest Ecosystems in Europe, vol. 388, 100–112. <http://dx.doi.org/10.1016/j.foreco.2016.10.027>, URL: <https://www.sciencedirect.com/science/article/pii/S0378112716307800>.
- Conedera, M., Wohlgemuth, T., Tanadini, M., Pezzatti, G.B., 2018. Drivers of broadleaved evergreen species spread into deciduous forests in the southern swiss alps. *Reg. Env. Chang.* 18 (2), 425–436. <http://dx.doi.org/10.1007/s10113-017-1212-7>.
- De Veaux, R.D., Ungar, L.H., 1994. Multicollinearity: A tale of two nonparametric regressions. In: *Selecting Models from Data: Artificial Intelligence and Statistics. IV*, Springer, pp. 393–402.
- Deshpande, P., Tripathi, S., Bhattacharya, A., 2024. Bayesian neural networks for satellite fog detection: Quantifying epistemic and aleatoric uncertainties. *Remote Sens. Earth Syst. Sci.* <http://dx.doi.org/10.1007/s41976-024-00155-7>.
- Dobrinčić, D., Gašparović, M., Medak, D., 2022. Evaluation of feature selection methods for vegetation mapping using multitemporal sentinel imagery. *Int. Arch. Photogramm. Remote Sens. Spat. Inf. Sci.* 43, 485–491, Publisher: Copernicus Publications Göttingen, Germany.
- Dubayah, R., Blair, J.B., Goetz, S., Fatoyinbo, L., Hansen, M., Healey, S., Hofton, M., Hurr, G., Kellner, J., Luthcke, S., Armstrong, J., Tang, H., Duncanson, L., Hancock, S., Jantz, P., Marselis, S., Patterson, P.L., Qi, W., Silva, C., 2020. The global ecosystem dynamics investigation: High-resolution laser ranging of the earth's forests and topography. *Sci. Remote Sens.* 1, 100002. <http://dx.doi.org/10.1016/j.srs.2020.100002>, URL: <https://www.sciencedirect.com/science/article/pii/S2666017220300018>.
- Fuller, A., Millard, K., Green, J.R., 2022. Transfer learning with pretrained remote sensing transformers. <http://dx.doi.org/10.48550/arXiv.2209.14969>, URL: <http://arxiv.org/abs/2209.14969>, arXiv:2209.14969 [cs].
- Gao, Y., Zhang, M., Li, W., Tao, R., 2024. Distribution-independent domain generalization for multisource remote sensing classification. *IEEE Trans. Neural Netw. Learn. Syst.* 1–12. <http://dx.doi.org/10.1109/TNNLS.2024.3490577>, Conference Name: IEEE Transactions on Neural Networks and Learning Systems URL: <https://ieeexplore.ieee.org/abstract/document/10750894>.
- Ge, S., Antropov, O., Häme, T., McRoberts, R.E., Miettinen, J., 2023. Deep learning model transfer in forest mapping using multi-source satellite SAR and optical images. *Remote Sens.* 15 (21), 5152. <http://dx.doi.org/10.3390/rs15215152>, Number: 21 Publisher: Multidisciplinary Digital Publishing Institute URL: <https://www.mdpi.com/2072-4292/15/21/5152>.
- Gitelson, A.A., 2004. Wide dynamic range vegetation index for remote quantification of biophysical characteristics of vegetation. *J. Plant Physiol.* 161 (2), 165–173, Publisher: Elsevier.
- Grabska, E., Hostert, P., Pflugmacher, D., Ostapowicz, K., 2019. Forest stand species mapping using the sentinel-2 time series. *Remote Sens.* 11 (10), 1197. <http://dx.doi.org/10.3390/rs11101197>, Number: 10 Publisher: Multidisciplinary Digital Publishing Institute URL: <https://www.mdpi.com/2072-4292/11/10/1197>.
- Guo, Y., Li, Z., Chen, E., Zhang, X., Zhao, L., Xu, E., Hou, Y., Sun, R., 2020. An end-to-end deep fusion model for mapping forests at tree species levels with high spatial resolution satellite imagery. *Remote Sens.* 12 (20), 3324. <http://dx.doi.org/10.3390/rs12203324>, Number: 20 Publisher: Multidisciplinary Digital Publishing Institute URL: <https://www.mdpi.com/2072-4292/12/20/3324>.
- Guo, X., Yu, H., 2022. On the domain adaptation and generalization of pretrained language models: A survey. <http://dx.doi.org/10.48550/arXiv.2211.03154>, URL: <http://arxiv.org/abs/2211.03154>, arXiv:2211.03154 [cs].
- Hamedianfar, A., Mohamedou, C., Kangas, A., Vauhkonen, J., 2022. Deep learning for forest inventory and planning: a critical review on the remote sensing approaches so far and prospects for further applications. *Forestry* 95 (4), 451–465, Publisher: Oxford University Press.
- Hemmerling, J., Pflugmacher, D., Hostert, P., 2021. Mapping temperate forest tree species using dense sentinel-2 time series. *Remote Sens. Environ.* 267, 112743. <http://dx.doi.org/10.1016/j.rse.2021.112743>, URL: <https://www.sciencedirect.com/science/article/pii/S0034425721004636>.
- Hendrycks, D., Lee, K., Mazeika, M., 2019. Using pre-training can improve model robustness and uncertainty. In: *International Conference on Machine Learning*, PMLR, pp. 2712–2721.
- Ho Tong Minh, D., Ienco, D., Gaetano, R., Lalande, N., Ndikumana, E., Osman, F., Maurel, P., 2018. Deep recurrent neural networks for winter vegetation quality mapping via multitemporal SAR sentinel-1. *IEEE Geosci. Remote Sens. Lett.* 15 (3), 464–468. <http://dx.doi.org/10.1109/LGRS.2018.2794581>, Conference Name: IEEE Geoscience and Remote Sensing Letters.
- Huete, A., Liu, H., Batchily, K., van Leeuwen, W., 1997. A comparison of vegetation indices over a global set of TM images for EOS-MODIS. *Remote Sens. Environ.* 59 (3), 440–451. [http://dx.doi.org/10.1016/S0034-4257\(96\)00112-5](http://dx.doi.org/10.1016/S0034-4257(96)00112-5), URL: <https://www.sciencedirect.com/science/article/pii/S0034425796001125>.
- Immitzer, M., Atzberger, C., Koukal, T., 2012. Tree species classification with random forest using very high spatial resolution 8-band WorldView-2 satellite data. *Remote Sens.* 4 (9), 2661–2693. <http://dx.doi.org/10.3390/rs4092661>, Number: 9 Publisher: Molecular Diversity Preservation International URL: <https://www.mdpi.com/2072-4292/4/9/2661>.
- Ismail Fawaz, H., Lucas, B., Forestier, G., Pelletier, C., Schmidt, D.F., Weber, J., Webb, G.I., Idoumghar, L., Muller, P.-A., Petitjean, F., 2020. Inceptiontime: Finding alexnet for time series classification. *Data Min. Knowl. Discov.* 34 (6), 1936–1962, Publisher: Springer.
- Jandt, U., Bruehlheide, H., Berg, C., Bernhardt-Römermann, M., Blüml, V., Bode, F., Dengler, J., Diekmann, M., Dierschke, H., Doerfler, I., Döring, U., Dullinger, S., Härdtle, W., Haider, S., Heinken, T., Horchler, P., Jansen, F., Kudernatsch, T., Kuhn, G., Lindner, M., Matesanz, S., Metz, K., Meyer, S., Müller, F., Müller, N., Naaf, T., Peppeler-Lisbach, C., Poschod, P., Roscher, C., Rosenthal, G., Rumpf, S.B., Schmidt, W., Schrautzer, J., Schwabe, A., Schwartze, P., Sperle, T., Stanik, N., Stroh, H.-G., Storm, C., Voigt, W., von Heßberg, A., von Oheimb, G., Wagner, E.-R., Wegener, U., Wesche, K., Wittig, B., Wulf, M., 2022. ReSurveyGermany: Vegetation-plot time-series over the past hundred years in Germany. *Sci. Data* 9 (1), 631. <http://dx.doi.org/10.1038/s41597-022-01688-6>, Publisher: Nature Publishing Group URL: <https://www.nature.com/articles/s41597-022-01688-6>.
- Kacic, P., Hirner, A., Da Ponte, E., 2021. Fusing sentinel-1 and -2 to model GEDI-derived vegetation structure characteristics in GEE for the paraguay chaco. *Remote Sens.* 13 (24), 5105. <http://dx.doi.org/10.3390/rs13245105>, Number: 24 Publisher: Multidisciplinary Digital Publishing Institute URL: <https://www.mdpi.com/2072-4292/13/24/5105>.
- Kattenborn, T., Leitloff, J., Schiefer, F., Hinz, S., 2021. Review on convolutional neural networks (CNN) in vegetation remote sensing. *ISPRS J. Photogramm. Remote Sens.* 173, 24–49, Publisher: Elsevier.
- Khanal, N., Matin, M.A., Uddin, K., Poortinga, A., Chishtie, F., Tenneson, K., Saah, D., 2020. A comparison of three temporal smoothing algorithms to improve land cover classification: a case study from NEPAL. *Remote Sens.* 12 (18), 2888, Publisher: MDPI.
- Kiefer, N., Weyrauch, A., Öz, M., Streit, A., Götz, M., Debus, C., 2024. A comparative study of pruning methods in transformer-based time series forecasting. <http://dx.doi.org/10.48550/arXiv.2412.12883>, URL: <http://arxiv.org/abs/2412.12883>, arXiv:2412.12883 [cs].
- Kollert, A., Bremer, M., Löw, M., Rutzinger, M., 2021. Exploring the potential of land surface phenology and seasonal cloud free composites of one year of sentinel-2 imagery for tree species mapping in a mountainous region. *Int. J. Appl. Earth Obs. Geoinf.* 94, 102208. <http://dx.doi.org/10.1016/j.jag.2020.102208>, URL: <https://www.sciencedirect.com/science/article/pii/S030324342030461X>.

- Lakshminarayanan, B., Pritzel, A., Blundell, C., 2017. Simple and scalable predictive uncertainty estimation using deep ensembles. *Adv. Neural Inf. Process. Syst.* 30.
- Lang, N., Jetz, W., Schindler, K., Wegner, J.D., 2023. A high-resolution canopy height model of the earth. *Nat. Ecol. Evol.* 7 (11), 1778–1789. <http://dx.doi.org/10.1038/s41559-023-02206-6>, Publisher: Nature Publishing Group URL: <https://www.nature.com/articles/s41559-023-02206-6>.
- Langanke, T., Herrmann, D., Ramminger, G., Buzzo, G., Berndt, F., 2018. Copernicus Land Monitoring Service - High Resolution Layer Forest. European Environment Agency (EEA).
- Li, H., Li, Y., Zhang, G., Liu, R., Huang, H., Zhu, Q., Tao, C., 2022a. Global and local contrastive self-supervised learning for semantic segmentation of HR remote sensing images. *IEEE Trans. Geosci. Remote Sens.* 60, 1–14. <http://dx.doi.org/10.1109/TGRS.2022.3147513>, URL: <http://arxiv.org/abs/2106.10605>, arXiv:2106.10605 [cs].
- Li, R., Xia, H., Zhao, X., Guo, Y., 2023. Mapping evergreen forests using new phenology index, time series sentinel-1/2 and google earth engine. *Ecol. Indic.* 149, 110157, Publisher: Elsevier.
- Li, H., Zhao, X., Li, S., Wan, F., Zhao, D., Zeng, J., 2022b. Improving molecular property prediction through a task similarity enhanced transfer learning strategy. *IScience* 25 (10), 105231. <http://dx.doi.org/10.1016/j.isci.2022.105231>, URL: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9579493/>.
- Liu, J., Shen, Z., He, Y., Zhang, X., Xu, R., Yu, H., Cui, P., 2023. Towards out-of-distribution generalization: A survey. <http://dx.doi.org/10.48550/arXiv.2108.13624>, URL: <http://arxiv.org/abs/2108.13624> [cs].
- Ma, Y., Chen, S., Ermon, S., Lobell, D.B., 2024. Transfer learning in environmental remote sensing. *Remote Sens. Environ.* 301, 113924, Publisher: Elsevier.
- Ma, Y., Zhang, Z., Kang, Y., Özdoğan, M., 2021. Corn yield prediction and uncertainty analysis based on remotely sensed variables using a Bayesian neural network approach. *Remote Sens. Environ.* 259, 112408. <http://dx.doi.org/10.1016/j.rse.2021.112408>, URL: <https://www.sciencedirect.com/science/article/pii/S0034425721001267>.
- Maaten, L.v.d., Hinton, G., 2008. Visualizing data using t-SNE. *J. Mach. Learn. Res.* 9 (86), 2579–2605, URL: <http://jmlr.org/papers/v9/vandemaaten08a.html>.
- Montero, D., Aybar, C., Mahecha, M.D., Martinuzzi, F., Söchting, M., Wieneke, S., 2023. A standardized catalogue of spectral indices to advance the use of remote sensing in earth system research. *Sci. Data* 10 (1), 197. <http://dx.doi.org/10.1038/s41597-023-02096-0>, Publisher: Nature Publishing Group URL: <https://www.nature.com/articles/s41597-023-02096-0>.
- Mucsányi, B., Kirchhof, M., Oh, S.J., 2025. Benchmarking uncertainty disentanglement: Specialized uncertainties for specialized tasks. *Adv. Neural Inf. Process. Syst.* 37, 50972–51038, URL: https://proceedings.neurips.cc/paper_files/paper/2024/hash/5afa9cb1e917b898ad418216dc726fdb-Abstract-Datasets_and_Benchmarks_Track.html.
- Muro, J., Linstädter, A., Magdon, P., Wöllauer, S., Männer, F.A., Schwarz, L.-M., Ghazaryan, G., Schultz, J., Malenovsky, Z., Dubovyk, O., 2022. Predicting plant biomass and species richness in temperate grasslands across regions, time, and land management with remote sensing and deep learning. *Remote Sens. Environ.* 282, 113262. <http://dx.doi.org/10.1016/j.rse.2022.113262>, URL: <https://www.sciencedirect.com/science/article/pii/S0034425722003686>.
- Nix, D.A., Weigend, A.S., 1994. Estimating the mean and variance of the target probability distribution. *ICNN'94, In: Proceedings of 1994 IEEE International Conference on Neural Networks*, vol. 1, IEEE, pp. 55–60.
- Oguiza, I., 2023. Tsai - a state-of-the-art deep learning library for time series and sequential data. URL: <https://github.com/timeseriesAI/tsai>.
- Peñuelas, J., Boada, M., 2003. A global change-induced biome shift in the montseny mountains (NE Spain). *Global Change Biol.* 9 (2), 131–140. <http://dx.doi.org/10.1046/j.1365-2486.2003.00566.x>.
- Pettorelli, N., Schulte to Bühne, H., Tulloch, A., Dubois, G., Macinnis-Ng, C., Queirós, A.M., Keith, D.A., Wegmann, M., Schrod, F., Stellmes, M., Sonnenschein, R., Geller, G.N., Roy, S., Somers, B., Murray, N., Bland, L., Geijzendorffer, I., Kerr, J.T., Broszeit, S., Leitão, P.J., Duncan, C., El Serafy, G., He, K.S., Blanchard, J.L., Lucas, R., Mairota, P., Webb, T.J., Nicholson, E., 2018. Satellite remote sensing of ecosystem functions: opportunities, challenges and way forward. *Remote Sens. Ecol. Conserv.* 4 (2), 71–93. <http://dx.doi.org/10.1002/rse2.59>, eprint: URL: <https://onlinelibrary.wiley.com/doi/abs/10.1002/rse2.59>.
- Pocivčiūtė, M., Eilertsen, G., Jarkman, S., Lundström, C., 2022. Generalisation effects of predictive uncertainty estimation in deep learning for digital pathology. *Sci. Rep.* 12 (1), 8329, Publisher: Nature Publishing Group UK London.
- Praticò, S., Solano, F., Di Fazio, S., Modica, G., 2021. Machine learning classification of mediterranean forest habitats in google earth engine based on seasonal sentinel-2 time-series and input image composition optimisation. *Remote Sens.* 13 (4), 586. <http://dx.doi.org/10.3390/rs13040586>, Number: 4 Publisher: Multidisciplinary Digital Publishing Institute URL: <https://www.mdpi.com/2072-4292/13/4/586>.
- Rather, I.H., Kumar, S., Gandomi, A.H., 2024. Breaking the data barrier: a review of deep learning techniques for democratizing AI with small datasets. *Artif. Intell. Rev.* 57 (9), 226. <http://dx.doi.org/10.1007/s10462-024-10859-3>.
- Sabatini, F.M., Lenoir, J., Hattab, T., Arnst, E.A., Chytrý, M., Dengler, J., De Ruffray, P., Hennekens, S.M., Jandt, U., Jansen, F., Jiménez-Alfaro, B., Kattge, J., Levesley, A., Pillar, V.D., Purschke, O., Sandel, B., Sultana, F., Aavik, T., Acici, S., Acosta, A.T.R., Aguilero, E., Alvarez, M., Apostolova, I., Arfin Khan, M.A.S., Arroyo, L., Attorre, F., Aubin, I., Banerjee, A., Batters, M., Bergeron, Y., Bergmeier, E., Biurru, I., Bjorkman, A.D., Bonari, G., Bondareva, V., Brunet, J., Čarni, A., Casella, L., Cayuela, L., Černý, T., Chepinoga, V., Csiky, J., Čuštrevska, R., De Bie, E., de Gasper, A.L., De Sanctis, M., Dimopoulos, P., Dolezal, J., Dziuba, T., El-Sheikh, M.A.E.-R.M., Enquist, B., Ewald, J., Fazayeli, F., Field, R., Finckh, M., Gachet, S., Galán-de Mera, A., Garbolino, E., Gholizadeh, H., Giorgis, M., Golub, V., Alsos, I.G., Grytnes, J.-A., Guerin, G.R., Gutiérrez, A.G., Haider, S., Hatim, M.Z., Hérault, B., Hinojos Mendoza, G., Hölzel, N., Homeier, J., Hubau, W., Indreica, A., Janssen, J.A.M., Jedrzejek, B., Jentsch, A., Jurgens, N., Kacki, Z., Kapfer, J., Karger, D.N., Kavgaci, A., Kearsley, E., Kessler, M., Khanina, L., Killeen, T., Korolyuk, A., Kreft, H., Kühn, H.S., Kuzemko, A., Landucci, F., Lengyel, A., Lens, F., Lingner, D.V., Liu, H., Lysenko, T., Mahecha, M.D., Marcenò, C., Martynenko, V., Moeslund, J.E., Monteagudo Mendoza, A., Mucina, L., Müller, J.V., Munzing, J., Naqinezhad, A., Noroozi, J., Nowak, A., Onyshchenko, V., Overbeck, G.E., Pärtel, M., Pauchard, A., Peet, R.K., Peñuelas, J., Pérez-Haase, A., Peterka, T., Petřík, P., Peyre, G., Phillips, O.L., Prokhorov, V., Rašomavičius, V., Revermann, R., Rivas-Torres, G., Rodwell, J.S., Ruprecht, E., Rüsina, S., Samimi, C., Schmidt, M., Schrod, F., Shan, H., Shirokikh, P., Šibík, J., Šilc, U., Sklenář, P., Skvorc, Z., Sparrow, B., Sperandii, M.G., Stanić, Z., Svenning, J.-C., Tang, Z., Tang, C.Q., Tsiropidis, I., Vanselow, K.A., Vázquez Martínez, R., Vassilev, K., Vélaz-Martín, E., Venanzoni, R., Vitrans, A.C., Violle, C., Virtanen, R., von Wehrden, H., Wagner, V., Walker, D.A., Waller, D.M., Wang, H.-F., Wesche, K., Whitfield, T.J.S., Willner, W., Wiser, S.K., Wohlgemuth, T., Yamalov, S., Zobel, M., Bruehlheide, H., 2021. SPlotOpen – an environmentally balanced, open-access, global dataset of vegetation plots. *Glob. Ecol. Biogeogr.* 30 (9), 1740–1764. <http://dx.doi.org/10.1111/geb.13346>, eprint: URL: <https://onlinelibrary.wiley.com/doi/abs/10.1111/geb.13346>.
- Safonova, A., Ghazaryan, G., Stiller, S., Main-Knorn, M., Nendel, C., Ryo, M., 2023. Ten deep learning techniques to address small data problems with remote sensing. *Int. J. Appl. Earth Obs. Geoinf.* 125, 103569. <http://dx.doi.org/10.1016/j.jag.2023.103569>, URL: <https://www.sciencedirect.com/science/article/pii/S156984322300393X>.
- Salazar, J.J., Garland, L., Ochoa, J., Pycr, M.J., 2022. Fair train-test split in machine learning: Mitigating spatial autocorrelation for improved prediction accuracy. *J. Pet. Sci. Eng.* 209, 109885. <http://dx.doi.org/10.1016/j.petrol.2021.109885>, URL: <https://www.sciencedirect.com/science/article/pii/S0920410521015023>.
- Schiefer, F., Kattenborn, T., Frick, A., Frey, J., Schall, P., Koch, B., Schmidtlein, S., 2020. Mapping forest tree species in high resolution UAV-based RGB-imagery by means of convolutional neural networks. *ISPRS J. Photogramm. Remote Sens.* 170, 205–215. <http://dx.doi.org/10.1016/j.isprsjprs.2020.10.015>, URL: <https://www.sciencedirect.com/science/article/pii/S0924646020302938>.
- Shao, Y., Lunetta, R.S., Wheeler, B., Iames, J.S., Campbell, J.B., 2016. An evaluation of time-series smoothing algorithms for land-cover classifications using MODIS-NDVI multi-temporal data. *Remote Sens. Environ.* 174, 258–265, Publisher: Elsevier.
- Simonyan, K., Vedaldi, A., Zisserman, A., 2014. Deep inside convolutional networks: Visualising image classification models and saliency maps. <http://dx.doi.org/10.48550/arXiv.1312.6034>, URL: <http://arxiv.org/abs/1312.6034>, arXiv:1312.6034 [cs].
- Smith, M.J., Fleming, L., Geach, J.E., 2024. Earthtpt: a time series foundation model for earth observation. <http://dx.doi.org/10.48550/arXiv.2309.07207>, URL: <http://arxiv.org/abs/2309.07207>, arXiv:2309.07207 [cs].
- Sylvain, J.-D., Drolet, G., Thiffault, E., Antcl, F., 2024. High-resolution mapping of tree species and associated uncertainty by combining aerial remote sensing data and convolutional neural networks ensemble. *Int. J. Appl. Earth Obs. Geoinf.* 131, 103960, Publisher: Elsevier.
- Tan, C.W., Bergmeir, C., Petitjean, F., Webb, G.I., 2021. Time series extrinsic regression: Predicting numeric values from time series data. *Data Min. Knowl. Discov.* 35 (3), 1032–1060, Publisher: Springer.
- Tarquini, S., Vinci, S., Favalli, M., Doumaz, F., Fornaciai, A., Nannipieri, L., 2012. Release of a 10-m-resolution DEM for the Italian territory: Comparison with global-coverage DEMs and anaglyph-mode exploration via the web. *Comput. Geosci.* 38 (1), 168–170. <http://dx.doi.org/10.1016/j.cageo.2011.04.018>, URL: <https://www.sciencedirect.com/science/article/pii/S0098300411001609>.
- Theunissen, M.W., Davel, M.H., Barnard, E., 2019. Insights regarding overfitting on noise in deep learning. In: *FAIR*. pp. 49–63.
- Tom, G., Hickman, R.J., Zinzowadia, A., Mohajeri, A., Sanchez-Lengeling, B., Aspuru-Guzik, A., 2023. Calibration and generalizability of probabilistic models on low-data chemical datasets with DIONYSUS. *Digit. Discov.* 2 (3), 759–774, Publisher: Royal Society of Chemistry.
- Tran, B.N., Van Der Kwast, J., Seyoum, S., Uijlenhoet, R., Jewitt, G., Mul, M., 2023. Uncertainty assessment of satellite remote-sensing-based evapotranspiration estimates: a systematic review of methods and gaps. *Hydrol. Earth Syst. Sci.* 27 (24), 4505–4528, Publisher: Copernicus Publications Göttingen, Germany.
- Tseng, G., Cartuyvels, R., Zvonkov, I., Purohit, M., Rolnick, D., Kerner, H., 2024. Lightweight, pre-trained transformers for remote sensing timeseries. <http://dx.doi.org/10.48550/arXiv.2304.14065>, URL: <http://arxiv.org/abs/2304.14065>, arXiv:2304.14065 [cs].

- Tuia, D., Volpi, M., Copa, L., Kanevski, M., Munoz-Mari, J., 2011. A survey of active learning algorithms for supervised remote sensing image classification. *IEEE J. Sel. Top. Signal Process.* 5 (3), 606–617. <http://dx.doi.org/10.1109/JSTSP.2011.2139193>, Conference Name: IEEE Journal of Selected Topics in Signal ProcessingURL: <https://ieeexplore.ieee.org/abstract/document/5742970>.
- Valdenegro-Toro, M., Mori, D.S., 2022. A deeper look into aleatoric and epistemic uncertainty disentanglement. In: 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops. CVPRW, IEEE, pp. 1508–1516.
- Wilson, E.H., Sader, S.A., 2002. Detection of forest harvest type using multiple dates of landsat TM imagery. *Remote Sens. Environ.* 80 (3), 385–396, Publisher: Elsevier.
- Yang, W., Kobayashi, H., Wang, C., Shen, M., Chen, J., Matsushita, B., Tang, Y., Kim, Y., Bret-Harte, M.S., Zona, D., Oechel, W., Kondoh, A., 2019. A semi-analytical snow-free vegetation index for improving estimation of plant phenology in tundra and grassland ecosystems. *Remote Sens. Environ.* 228, 31–44. <http://dx.doi.org/10.1016/j.rse.2019.03.028>, URL: <https://www.sciencedirect.com/science/article/pii/S003442571930121X>.
- Yang, S., Yee, K., 2024. Towards reliable uncertainty quantification via deep ensemble in multi-output regression task. *Eng. Appl. Artif. Intell.* 132, 107871. <http://dx.doi.org/10.1016/j.engappai.2024.107871>, URL: <https://www.sciencedirect.com/science/article/pii/S0952197624000290>.
- Yosinski, J., Clune, J., Bengio, Y., Lipson, H., 2014. How transferable are features in deep neural networks?. <http://dx.doi.org/10.48550/arXiv.1411.1792>, URL: <http://arxiv.org/abs/1411.1792>, arXiv:1411.1792 [cs].
- Zerveas, G., Jayaraman, S., Patel, D., Bhamidipaty, A., Eickhoff, C., 2020. A transformer-based framework for multivariate time series representation learning. <http://dx.doi.org/10.48550/arXiv.2010.02803>, URL: <http://arxiv.org/abs/2010.02803>, arXiv:2010.02803 [cs].
- Zhang, C., Bengio, S., Hardt, M., Recht, B., Vinyals, O., 2021. Understanding deep learning (still) requires rethinking generalization. *Commun. ACM* 64 (3), 107–115, Publisher: ACM New York, NY, USA.